Article

Advancing subseasonal forecasting with machine learning

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Decision-makers rely on weather forecasts to plant crops, manage wildfires, allocate water and energy and prepare for weather extremes[1–3]. Today, such forecasts enjoy unprecedented accuracy as far as 2 weeks thanks to steady advances in physics-based dynamical models and data-driven artificial intelligence models[4–10]. However, model skill drops precipitously at subseasonal timescales (2–6 weeks ahead) due to compounding errors, systemic model biases and the chaotic nature of the atmosphere[11–15]. To counter this degradation, we introduce probabilistic bias correction, a machine learning framework that substantially reduces systematic error by learning to correct historical probabilistic forecasts. When applied to the leading dynamical and artificial intelligence models from the European Centre for Medium-Range Weather Forecasts (ECMWF), probabilistic bias correction doubles the modest subseasonal skill of the AI Forecasting System[16] and improves the skill of the operationally debiased dynamical model[17] for 91% of pressure, 92% of temperature and 98% of precipitation targets. We designed probabilistic bias correction for operational deployment, and, in the 2025 real-time forecasting competition of ECMWF[18], its global forecasts were the best for all weather variables and lead times, outperforming the dynamical models from six operational forecasting centres, an international dynamical multimodel ensemble, the AI Forecasting System of ECMWF and the forecasting systems of 34 teams worldwide. These probabilistic skill gains translate into more accurate prediction of extreme events and have the potential to improve agricultural planning, energy management and disaster preparedness in vulnerable communities[19].

Subseasonal forecasts—weather predictions 2–6 weeks ahead—lie at the heart of agricultural planning, water allocation, disaster preparedness and energy management[1–3]. Yet accurate prediction at this timescale remains difficult. Today, our best estimates of future weather come from dynamical models that numerically simulate how the atmosphere and oceans change over time[4,5] and artificial intelligence (AI) models that learn to predict the weather from historical data[6–8]. Because of the steady improvements in modelling, computation and data collection over the past half-century, deterministic forecasting systems can now provide accurate point estimates of the weather 1–2 weeks in advance[10]. However, the 2-week horizon also marks a transition in predictability: beyond this point, even small differences in initial weather conditions can lead to dramatic differences in outcomes and a single deterministic forecast cannot capture this uncertainty[20].

To surmount this predictability barrier, one often turns to probabilistic forecasting, that is, predicting a distribution over future weather rather than a single point estimate[21–23]. Probabilistic forecasting explicitly encodes forecast uncertainty and provides actionable estimates of the likelihood of high-impact extreme events. For both dynamical and AI models, probabilistic forecasts are typically generated by perturbing the inputs, parameters or internal state of a model to produce an ensemble of possible future weather states[13,23–25]. Although these ensembles are routinely skilful at shorter lead times (to 15 days), operational dynamical ensembles and even state-of-the-art probabilistic AI routinely underperform the climatological average at longer subseasonal lead times, especially outside the tropics[13].

A core difficulty is that leading subseasonal ensembles are generated iteratively over many time steps and even small per-step modelling errors compound into large systematic errors over several weeks[14]. This effect is especially pronounced for precipitation, which is sensitive to cascading inaccuracies across model components and highly variable in space and time[4,26]. Indeed, subseasonal forecasting has long been considered a 'predictability desert' because of the compounding model errors and complex interdependencies among the atmosphere, land, ocean and sea-ice components that evolve with different timescales[11,15].

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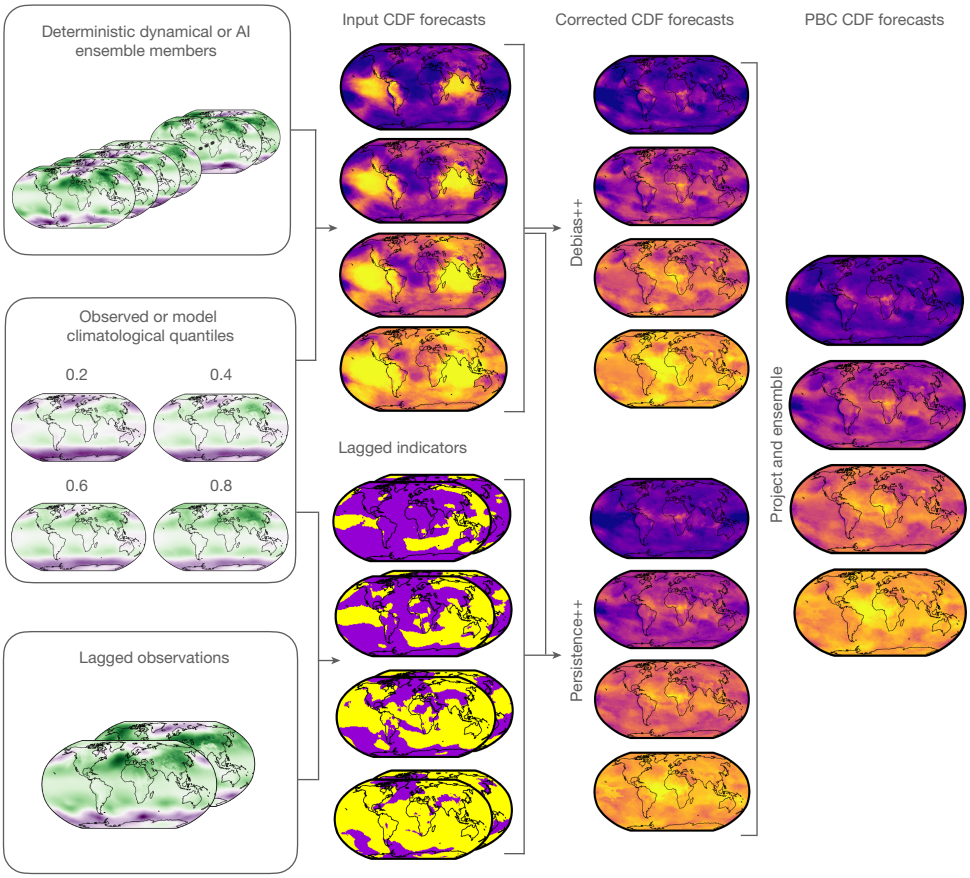

Fig. 1 | PBC schematic. PBC takes as input lagged observations, climatological quantiles (for example, quintiles, terciles or deciles) and the deterministic forecasts of any dynamical or AI model and outputs a learning-enhanced probabilistic forecast designed to correct systematic prediction errors. Coastlines from Natural Earth (https://www.naturalearthdata.com/).

Although existing subseasonal ensembles are indeed error-prone, we show that a component of their systematic error is predictable and correctable using machine learning, allowing significant gains in skill with minimal cost. To achieve this, we introduce a probabilistic bias correction (PBC) framework that adaptively learns to correct the forecast distribution of a model using its own historical predictions and observations of past weather. PBC is compatible with any input model and we use it to advance the state of the art in both data-driven and hybrid (AI + dynamical) subseasonal forecasting. The resulting forecasts are designed for operational deployment and, in real-time competition[18], outperform the best alternative dynamical, AI and hybrid forecasting systems from forecasting centres and competitors around the globe.

Probabilistic bias correction

PBC is a new machine learning framework for improving subseasonal forecasts from dynamical, AI or hybrid prediction systems. Given a raw ensemble of deterministic forecasts, PBC outputs learning-enhanced probabilities designed to correct systematic errors while preserving the predictive signal in the underlying model. Our approach is inspired by the deterministic bias correction framework of ref. 27 but operates directly on the space of probability distributions rather than at the level of raw observations. The added flexibility allows us to directly optimize not only the centre of a distribution but also its spread and shape.

Figure 1 illustrates a sample application of PBC. First, an input ensemble is converted into an initial probabilistic forecast compared with climatological quantile bins derived from historical observations or from the hindcasts of the model. Next, two computationally efficient machine learning models—Debias++ and Persistence++—are applied in parallel to produce complementary corrections of the initial forecast. Debias++ perturbs the forecast using location-, date- and quantile-specific corrections learned by minimizing probabilistic forecasting error over adaptively selected training periods. Meanwhile, Persistence++ blends the forecast with lagged observations and climatology to account for recent weather trends and day-of-year effects. Finally, the two corrections are projected onto the space of valid cumulative distribution functions (CDFs) and averaged to form the final PBC prediction. The Methods provide further details.

This framework can be used to improve forecasts for any weather variable, any spatial grid and any number of quantile bins. For concreteness, we adopt the following conventions of the AI Weather Quest subseasonal forecasting competition[18] run by the ECMWF. First, we focus on forecasting average 2-m temperature (over land), total precipitation (over land) and average mean sea-level pressure on a global 1.5° latitude–longitude grid with five quintile bins. Second, we evaluate subseasonal forecasts for week 3 (days 19–25) and week 4 (days 26–32) using ranked probability skill score (RPSS)[28], a standard measure of forecast improvement compared with a climatological baseline. Further details on the forecasting targets and evaluation are described in the Methods.

To assess the operational utility and versatility of PBC, we apply the framework to three state-of-the-art subseasonal forecasting models: the industry-leading dynamical ensemble from ECMWF[17], subseasonal AI Forecasting System (AIFS-SUBS) from ECMWF[16] and post-processing ensembles with transformers (POET), a hybrid model that postprocesses the ECMWF dynamical ensemble with convolutional and transformer neural networks[29]. As our measure of ground truth, we

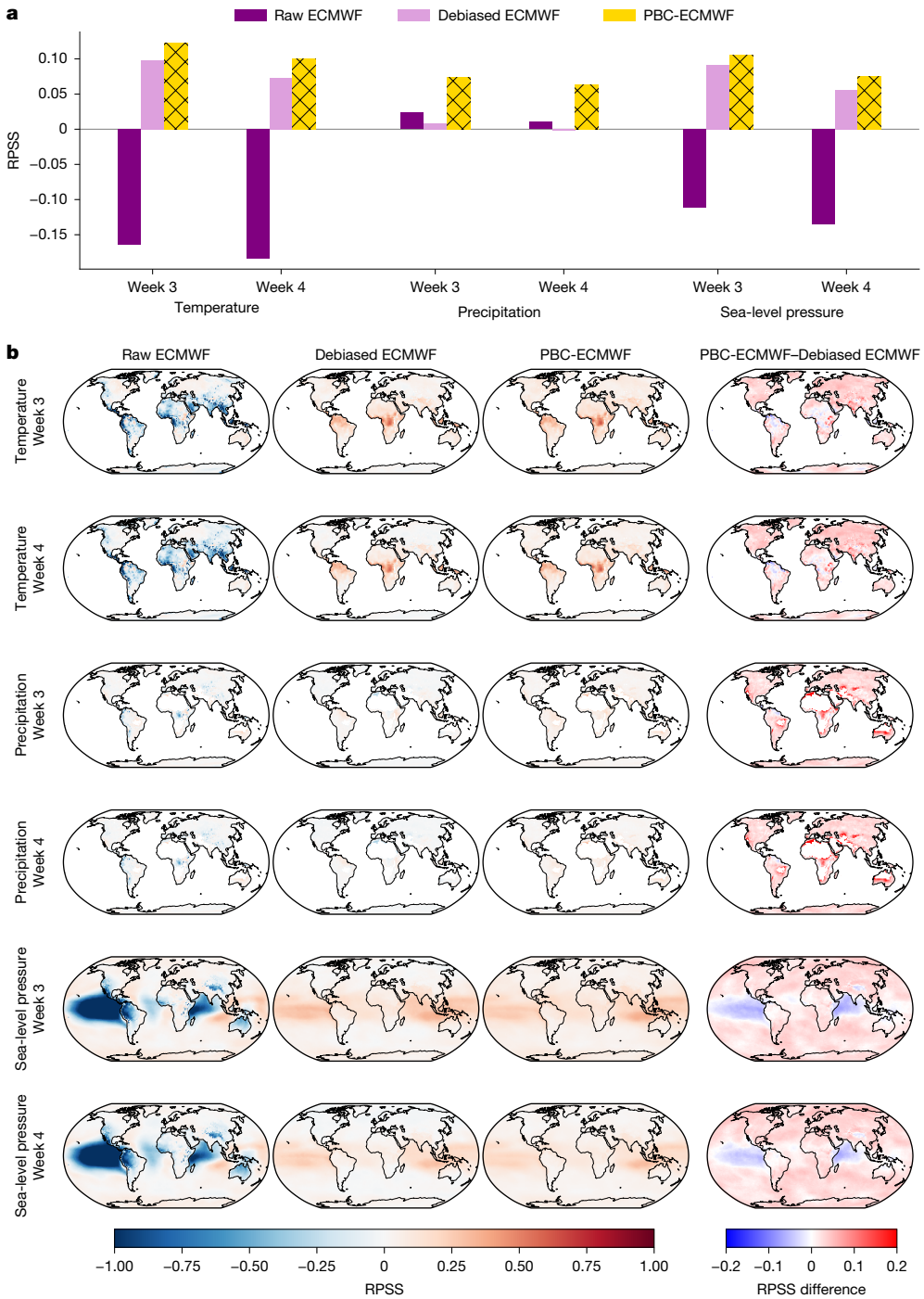

Fig. 2 | Global forecast skill of ECMWF and its correction. a, Across the years 2016–2024, PBC boosts the subseasonal skill of raw ECMWF forecasts more effectively than operational debiasing, transforming low- and no-skill inputs into skilful outputs that consistently outperform climatology. Negative RPSS indicates predictive performance worse than climatology, positive RPSS indicates improved performance and cross-hatching indicates significant improvement (paired one-sided stationary block bootstrap $P < 0.05$ for 239 target dates) compared with raw and debiased ECMWF. **b**, PBC improvements are broadly distributed with 98% of grid cells showing skill gains compared with debiased ECMWF for precipitation, 91–92% for temperature and 89–91% for sea-level pressure. Arid grid cells with no climatological precipitation are excluded. Coastlines in **b** from Natural Earth (https://www.naturalearthdata.com/).

adopt the ERA5 reanalysis dataset[30], which assimilates a wide range of observations to produce a global gridded record of the Earth's weather from 1940 to the present, but any other target for correction could be substituted. Because PBC is so lightweight, we also incrementally retrain it each day, using only the reanalysis and model forecast data that would have been observable on each forecast issuance date to mimic operational deployment (Methods).

Correcting dynamical forecasts with PBC

Figure 2 summarizes the pronounced skill gains obtained when PBC is applied to the dynamical ensemble from ECMWF, widely considered to be the most accurate weather model in the world and the gold standard for benchmarking subseasonal skill[31,32]. For each target variable and lead time, Fig. 2a reports global RPSS over the years 2016–2024,

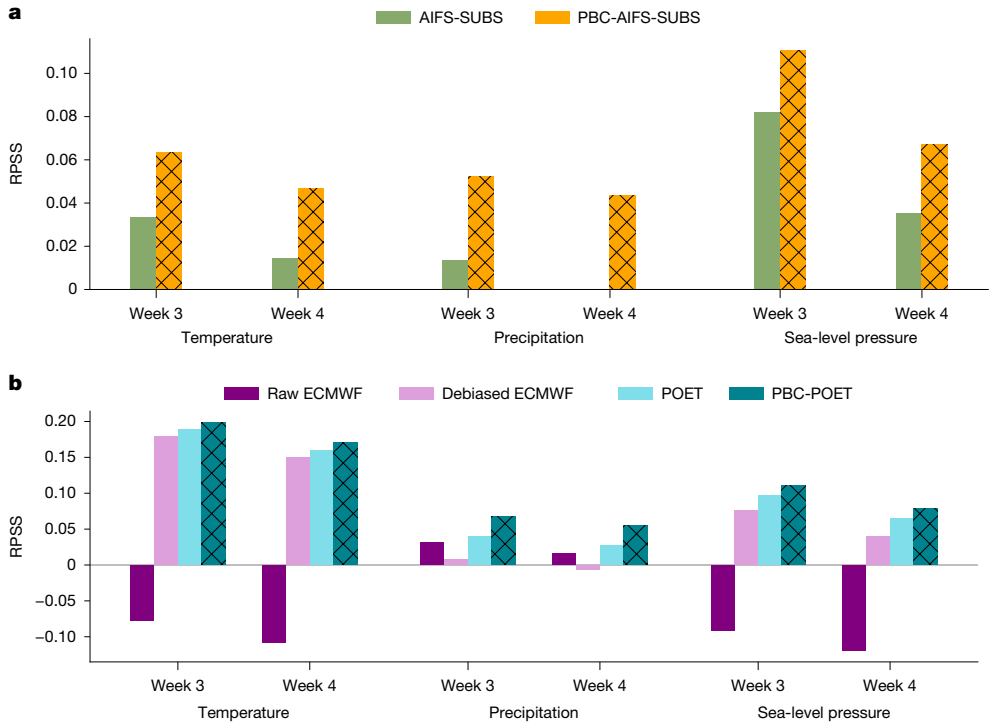

Fig. 3 | Global forecast skill of AI models and their corrections. a, In 2025, PBC consistently enhances the subseasonal skill of ECMWF AIFS-SUBS, doubling its average skill (0.03) across all variables and lead times. **b**, In 2024, the hybrid (AI + dynamical) model POET already improves the skill of its dynamical input, ECMWF. Nevertheless, PBC further enhances skill for every variable and lead time with gains of 68–98% for precipitation (compared with baseline skills of 0.03–0.04), 15–23% for mean sea-level pressure (compared with baseline skills of 0.08–0.10) and 5–7% for temperature (compared with baseline skills of 0.16–0.19). Cross-hatching indicates significant improvement (paired one-sided stationary block bootstrap $P < 0.05$ for (**a**) 179 and (**b**) 104 week 3 and 107 week 4 target dates) compared with all baselines.

whereas Fig. 2b shows the spatial RPSS distribution over the same time period. Negative RPSS indicates worse predictive performance than the climatological baseline, whereas positive RPSS indicates improved performance. PBC consistently improves the low-scoring raw ECMWF ensemble, tripling its positive precipitation skill (0.023 RPSS) and converting its negative-skill temperature and mean sea-level pressure forecasts into skilful predictions that consistently outperform climatology (Fig. 2a). Analogous gains are observed when skill is stratified by season (Extended Data Fig. 1) or by region (Extended Data Fig. 2) and the improvements are accompanied by a pronounced reduction in systematic model bias across the globe (Extended Data Figs. 3, 4 and 5).

To dampen the systemic effects of model bias, forecasting centres routinely debias their operational forecasts by comparing each deterministic prediction to quantiles derived from model hindcasts rather than observations[33]. Given the same deterministic predictions as input, PBC consistently outperforms this operational debiasing protocol, replacing the near-zero precipitation skill of debiased ECMWF with consistently positive RPSS and boosting the positive temperature and pressure RPSS of debiased ECMWF by 26–37% and 16–32% (above baseline skills of 0.07–0.10 and 0.06–0.11) respectively (Fig. 2a). The PBC gains compared with debiased ECMWF are also broadly distributed with 98% of grid cells showing skill gains for precipitation, 91–92% for temperature and 89–91% for sea-level pressure (Fig. 2b). These results demonstrate that PBC can transform unskilful inputs into reliably skilful outputs and serve as an improved drop-in replacement for existing operational debiasing protocols.

In fact, PBC can boost the skill of its dynamical input beyond the levels of existing AI-based subseasonal models. For example, ref. 13 recently introduced FuXi-S2S, a deep learning model that outperforms the dynamical ensemble of ECMWF at subseasonal lead times. Across the globe and the test years 2017–2021, PBC-ECMWF shows a pronounced increase in skill compared with FuXi-S2S for each variable and lead time, with 99–100% of grid cells showing skill gains for temperature, 97–99% for precipitation and 96–100% for mean sea-level pressure (Extended Data Fig. 6).

Decomposing PBC into its component parts (Extended Data Fig. 7a) yields four more insights into its skill gains. First, Debias++ and Persistence++ each significantly improve ($P < 0.05$) on raw ECMWF for all targets. Second, Persistence++ alone significantly improves ($P < 0.05$) on debiased ECMWF for each target except week-4 pressure, whereas Debias++ provides smaller improvements. Third, projection minimally affects skill as Debias++ and Persistence++ forecasts are typically valid CDFs. Finally, ensembling the complementary Persistence++ and Debias++ forecasts offers robust improvements across all variables and lead times.

Correcting AI forecasts with PBC

PBC can also be used to improve the skill of AI models. For example, Fig. 3a shows the substantial skill gains obtained when PBC is applied to the state-of-the-art subseasonal AIFS-SUBS of ECMWF. The AIFS is a deep learning model that exhibits greater subseasonal skill than debiased ECMWF for the challenging task of precipitation forecasting and comparable skill for temperature and mean sea-level pressure prediction[16]. Across the globe and the 2025 test year, PBC consistently amplifies AIFS-SUBS skill, doubling its mean RPSS (0.03) across all variables and lead times.

We also apply PBC to a hybrid (AI + dynamical) forecasting system, POET. POET takes as input a dynamical ensemble (in this case, ECMWF) and outputs a new ensemble with all members simultaneously corrected by a transformer embedded in a convolutional neural network[29]. POET serves as challenging input for PBC as its ensemble has already been corrected by AI and is substantially more skilful than its raw ECMWF input. Moreover, POET has been shown to outperform

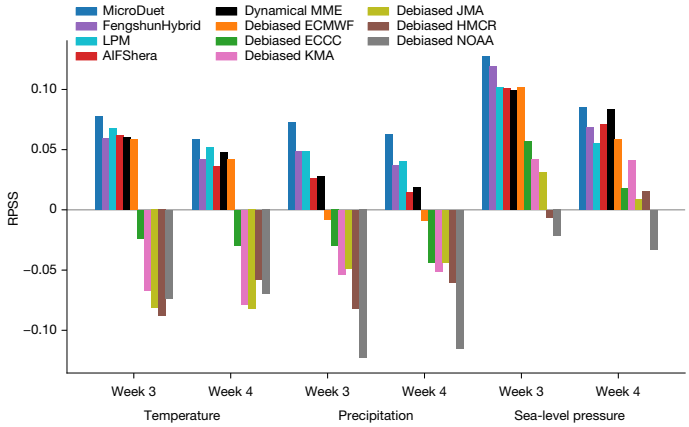

Fig. 4 | AI Weather Quest real-time global forecast skill. MicroDuet, an ensemble of PBC-ECMWF and PBC-POET, did best in the 2025 September–October–November AI Weather Quest competition for every variable and lead time, outperforming the debiased dynamical models of six government agencies (ECMWF, NOAA, ECCC, HMCR, KMA and JMA), their dynamical multimodel ensemble (Dynamical MME), three ECMWF AI models (including their top-performing AIFShera model) and the forecasting systems of 34 teams worldwide (including the top competitors, FengshunHybrid and LPM).

other statistical and AI-based post-processing techniques including member-by-member model output statistics[34,35], ensemble model output statistics[36], LeNet neural network post-processing[37] and pure transformer-based post-processing[38]. Nevertheless, PBC enhances POET skill for every variable and lead time with RPSS gains of 68–98% for precipitation (compared with baseline skills of 0.03–0.04), 15–23% for mean sea-level pressure (compared with baseline skills of 0.08–0.10) and 5–7% for temperature (compared with baseline skills of 0.16–0.19) in 2024 (Fig. 3b).

A decomposition of PBC into its component parts (Extended Data Fig. 7b) shows three distinct features contributing to its improvements. First, Debias++ alone improves on its POET input for each task, highlighting the value of direct CDF optimization compared with correction only in deterministic forecast space[27,29,34,35,38]. Second, Persistence++ delivers even greater skill for precipitation, highlighting the value of integrating recent observations compared with a correction that ignores this source of predictability[29,34–36,38,39]. Third, ensembling the two complementary corrections robustly captures the strengths of each, highlighting the value of model merging compared with monolithic correction[29,34–39].

AI Weather Quest forecasting competition

In 2025, ECMWF launched the AI Weather Quest, an international competition to benchmark and improve the state of operational subseasonal forecasting[18]. Each week, contestants submit real-time probabilistic forecasts for weather around the globe and, at the end of each season, models with the highest skills for week 3 and week 4 are recognized as winners. To benchmark the real-time operational utility of PBC, we entered an ensemble of PBC-ECMWF and PBC-POET into the competition under the name MicroDuet. In the inaugural September–October–November season, MicroDuet did the best for all weather variables and lead times, outperforming the debiased dynamical forecasts of six operational forecasting centres (ECMWF, NOAA, ECCC, HMCR, KMA and JMA)[40], a multimodel ensemble of these six dynamical forecasts, three versions of AIFS-SUBS and the forecasting systems of 34 teams worldwide (Fig. 4). MicroDuet was also the top-performing model in both forecast periods over the entire year.

The MicroDuet skill gains are especially pronounced for precipitation, as the debiased models of all six operational forecasting centres exhibit negative skill, performing worse than the climatological average. The Dynamical MME recovers positive skill by aggregating across all six forecasting centres, but, with skills of 0.06–0.07, MicroDuet alone is twice as skilful in week 3 and thrice as skilful in week 4. MicroDuet also did best for both lead times in the December–January–February season of the Quest and, across 37 teams, was the only model to outperform Dynamical MME for all variables[41]. Across both seasons, MicroDuet was also recognized as the only submission to significantly outperform debiased ECMWF ($P < 0.05$) for all variables and lead times[42].

In November 2025, the week-4 temperature forecast in MicroDuet of a mid-December cold air outbreak in the Eastern USA was shared in real-time[43] and verified as an extreme cold event[44] that was missed by debiased ECMWF (Extended Data Fig. 8). This demonstrates the potential of PBC to improve not only general forecast quality but also the early detection of impactful extreme-weather events. We next assess the skill of PBC when trained to target such extreme events.

Forecasting extreme events

The early identification of extreme weather is an important application of subseasonal forecasting that supports improved preparedness for hazards such as floods, heatwaves, droughts, storms and cold snaps. Figure 5 summarizes the substantial gains in extreme forecasting skill when PBC is applied to the dynamical ensemble from ECMWF. Following previous work[45], we define extreme highs as events verifying above the 95th percentile and extreme lows as events verifying below the 5th percentile. We then train PBC to target semidecile (5% quantile) increments rather than quintiles and measure skill using the Brier skill score (BSS)[28], a standard measure of improvement compared with a climatological baseline for a binary event.

For each target variable and extreme, Fig. 5a reports week-3 global BSS over the years 2016–2024, whereas Fig. 5b shows the spatial BSS distribution over the same time period. For extreme precipitation and all extreme lows, PBC again transforms its low- and no-skill inputs into skilful outputs that consistently outperform climatology. For extreme highs, PBC boosts the forecasting skill of operationally debiased ECMWF by 51% (compared with a baseline skill of 0.07) for temperature and 39% (compared with a baseline skill of 0.05) for pressure (Fig. 5a). The PBC improvements compared with debiased ECMWF are also broadly distributed with 96% of grid cells showing skill gains for precipitation, 89% for sea-level pressure and 85% for temperature (Fig. 5b). Comparable improvements are also observed in week 4 (Extended Data Fig. 9).

To evaluate whether these gains translate into improved hazard prediction, Fig. 5c shows flood forecasting skill across all flood events catalogued by the Global Disaster Alert and Coordination System (GDACS)[46] in the years 2016–2026. For each event, flood forecasting skill is quantified by the BSS for the top precipitation semidecile across the 20° × 20° bounding box centred on the flood centroid. Given the same raw ECMWF forecasts as input, PBC boosts flood forecast skill by 76–92%, whereas debiased ECMWF dramatically reduces it, driving 0.04–0.06 baseline skills to near zero. GDACS additionally categorizes each flood on the basis of its humanitarian impact and labels a flood severe (orange or red) if it results in more than 100 deaths or 80,000 displacements[46]. For these high-impact events, operational debiasing only degrades or matches the 0.04–0.07 BSS of raw ECMWF, whereas PBC enhances its severe flood forecasting skill by 86–150%.

Discussion

Our results indicate that probabilistic post-processing with machine learning can substantially boost the quality and usefulness of subseasonal forecasts. The PBC framework achieves this through four key properties. First, across the globe, PBC consistently improves the dynamical model from ECMWF, elevating hybrid (AI + dynamical) subseasonal forecasting. Second, PBC consistently improves leading subseasonal AI models, elevating fully data-driven subseasonal forecasting. Third,

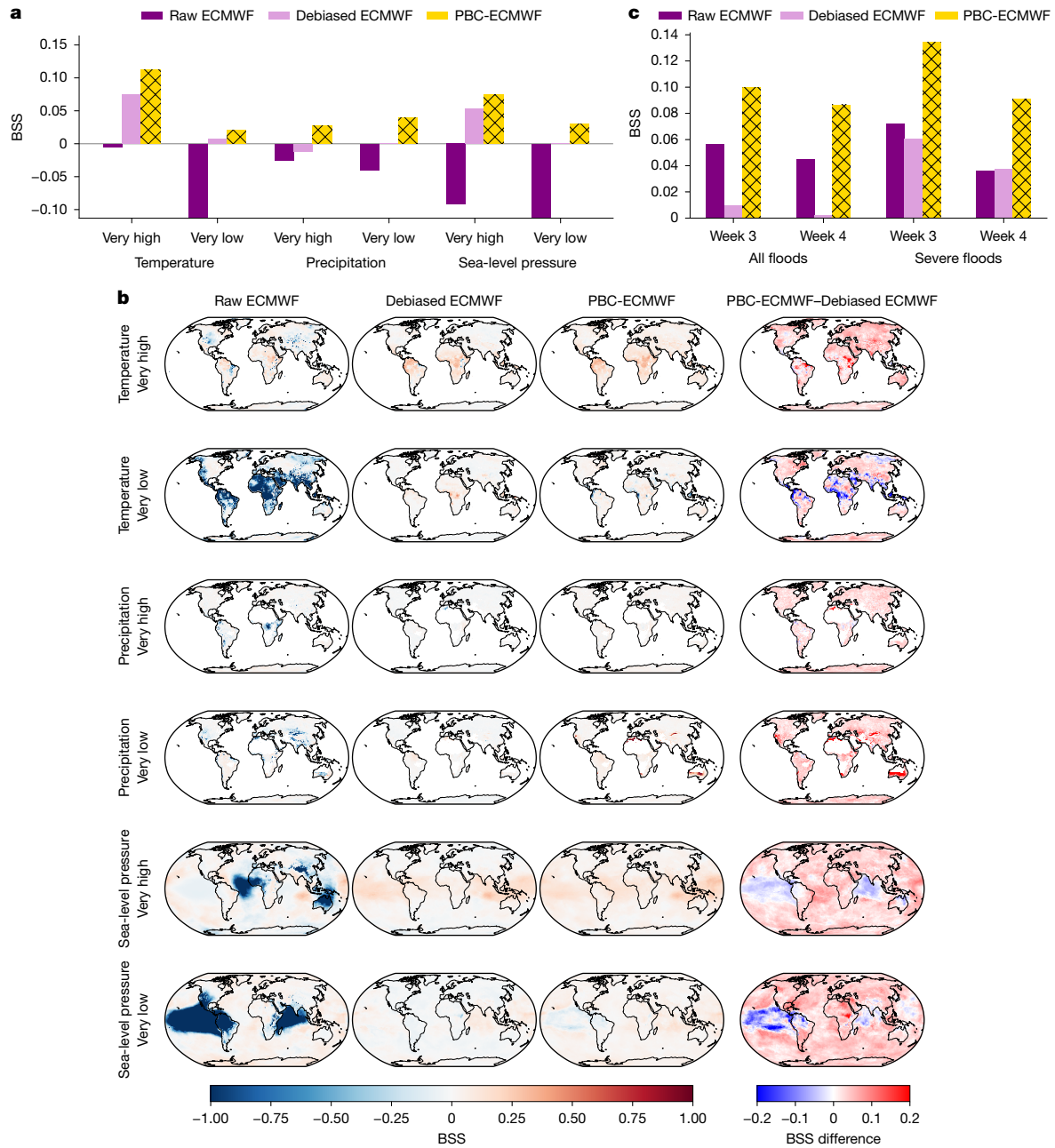

Fig. 5 | Forecasting extreme weather with ECMWF and itscorrection. **a**,**b**, Across the globe and the years 2016–2024, PBC boosts ECMWF week-3 extreme forecast skill more effectively than operational debiasing with 96% of grid cells showing skill gains for precipitation, 89% for sea-level pressure and 85% for temperature. Arid grid cells with no climatological precipitation are excluded. **c**, Across all 2016–2026 flood events catalogued by GDACS and severe GDACS floods with more than 100 deaths or 80,000 individuals displaced, operational debiasing routinely reduces raw ECMWF flood forecast skill, whereas PBC delivers pronounced improvements. Cross-hatching indicates significant improvement (**a**, paired one-sided stationary block bootstrap $P < 0.05$ for 939 target dates; **c**, paired one-sided bootstrap $P < 0.05$ for 88 severe and 2,435 total flood events) compared with raw and debiased ECMWF. Coastlines in **b** from Natural Earth (https://www.naturalearthdata.com/).

PBC forecasts are suitable for operational deployment and, in real-time competition, consistently outperform the best alternative dynamical, AI and hybrid forecasting systems worldwide. Fourth, PBC consistently improves the early detection of extreme and high-impact weather events. As climate change is predicted to increase the spatial and temporal occurrence of extreme weather, improvements in early warning capability are likely to reduce preventable morbidity and mortality[19].

PBC targets predictable errors rather than attempting to relearn full atmospheric dynamics. Hence, it should be viewed as an effective post-processing tool rather than a replacement for input model improvement. Fortunately, the PBC framework is also inherently adaptive: as dynamical and AI models are upgraded with improved physical representation, process representations, computational capacity or training data, PBC can be retrained to ingest and correct these higher-quality inputs. Because PBC is also inexpensive compared with the generation of large-scale dynamical or AI ensembles, the framework provides a low-overhead strategy for meteorological centres, including those in resource-constrained settings, to improve their existing forecast products.

With its subseasonal lead times, PBC also targets a critical gap in the weather forecasting spectrum: at longer leads, many high-impact events are not yet predictable, whereas, at shorter leads, many warnings

are no longer actionable. For example, long-lead seasonal forecasts failed to predict the disastrous 2018 Kenyan floods, but subseasonal rainfall forecasts provided a clear warning so that preparation was still possible[47]. Growing body of evidence demonstrates the value of skilful subseasonal forecasts for disaster response, energy production, healthcare and agriculture[1–3], especially in semi-arid regions in Eastern and Southern Africa, where agriculture remains highly susceptible to out-of-season rains[48,49]. However, this evidence often takes the form of isolated case studies leveraging custom post-processing to extract signal from raw model outputs[50]. PBC represents an important advance for subseasonal forecasting as its low-cost skill gains generalize across target variables, input models and regions. Through the accompanying release of our open-source code, we aim to provide a standardized tool for the community to bridge the gap between raw forecasts and actionable intelligence at subseasonal horizons and beyond.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at https://doi.org/10.1038/s41586-026-11126-x.

1. White, C. J. et al. Potential applications of subseasonal-to-seasonal (S2S) predictions. *Meteorol. Appl.* **24**, 315–325 https://doi.org/10.1002/met.1654 (2017).
2. Vitart, F. & Robertson, A. W. The sub-seasonal to seasonal prediction project (S2S) and the prediction of extreme events. *npj Clim. Atmos. Sci.* **1**, 3 https://doi.org/10.1038/s41612-018-0013-0 (2018).
3. Merryfield, W. J. et al. Current and emerging developments in subseasonal to decadal prediction. *Bull. Am. Meteorol. Soc.* **101**, 869–896 https://doi.org/10.1175/BAMS-D-19-0037.1 (2020).
4. Bauer, P., Thorpe, A. & Brunet, G. The quiet revolution of numerical weather prediction. *Nature* **525**, 47–55 (2015).
5. Ben Bouallègue, Z. et al. The rise of data-driven weather forecasting: a first statistical assessment of machine learning-based weather forecasts in an operational-like context. *Bull. Am. Meteorol. Soc.* **105**, 864–883 https://doi.org/10.1175/BAMS-D-23-0162.1 (2024).
6. Bi, K. et al. Accurate medium-range global weather forecasting with 3D neural networks. *Nature* **619**, 533–538 (2023).
7. Lam, R. et al. Learning skillful medium-range global weather forecasting. *Science* **382**, 1416–1421 https://doi.org/10.1126/science.adi2336 (2023).
8. Bodnar, C. et al. A foundation model for the Earth system. *Nature* **641**, 1180–1187 https://doi.org/10.1038/s41586-025-09005-y (2025).
9. Chen, L. et al. FuXi: a cascade machine learning forecasting system for 15-day global weather forecast. *npj Clim. Atmos. Sci.* **6**, 190 (2023).
10. Rasp, S. et al. Weatherbench 2: a benchmark for the next generation of data-driven global weather models. *J. Adv. Model. Earth Syst.* **16**, e2023MS004019 (2024).
11. Vitart, F., Robertson, A. W. & Anderson, D. L. Subseasonal to seasonal prediction project: bridging the gap between weather and climate. *Bull. World Meteorol. Org.* **61**, 23 (2012).
12. Cohen, J. et al. S2S reboot: an argument for greater inclusion of machine learning in subseasonal to seasonal forecasts. *WIREs Clim. Change* **10**, 00567 https://doi.org/10.1002/wcc.567 (2019).
13. Chen, L. et al. A machine learning model that outperforms conventional global subseasonal forecast models. *Nat. Commun.* **15**, 6425 https://doi.org/10.1038/s41467-024-50714-1 (2024).
14. Nathaniel, J. et al. Chaosbench: a multi-channel, physics-based benchmark for subseasonal-to-seasonal climate prediction. *Adv. Neural Info. Process. Syst.* **37**, 43715–43729 (2024).
15. Richter, J.H. & Joseph, E. Scientists must join forces to solve forecasting's predictability desert. *EOS* https://doi.org/10.1029/2025EO250389 (2025).
16. Schloer, J. et al. AIFS-SUBS: extending data-driven forecasting to sub-seasonal timescales. Preprint at https://doi.org/10.48550/arXiv.2607.05100 (2026).
17. Tuppi, L., Ekblom, M., Ollinaho, P. & Järvinen, H. Simultaneous optimization of 20 key parameters of the integrated forecasting system of ECMWF using OpenIFS. Part I: effect on deterministic forecasts. *Mon. Weather Rev.* **151**, 1325–1346 (2023).
18. Loegel, O., Talib, J., Vitart, F., Hoffmann, J. & Chantry, M. The AI Weather Quest: an international competition for sub-seasonal forecasting with AI. *Mach. Learn. Earth* **1**, 010701 https://doi.org/10.1088/3049-4753/adf649 (2025).
19. Shrader, J. G., Thies, S., Bakkensen, L., Linsenmeier, M. & Lemoine, D. Weather forecasts become more important for reducing mortality as the climate warms. *Proc. Natl Acad. Sci. USA* **123**, 2523372123 https://doi.org/10.1073/pnas.2523372123 (2026).
20. Lorenz, E. N. The predictability of a flow which possesses many scales of motion. *Tellus* **21**, 289–307 (1969).
21. Gneiting, T. & Raftery, A. E. Weather forecasting with ensemble methods. *Science* **310**, 248–249 (2005).
22. Palmer, T. N. The economic value of ensemble forecasts as a tool for risk assessment: from days to decades. *Q. J. R. Meteorol. Soc.* **128**, 747–774 (2002).
23. Mylne, K. Probability forecasts—part 1: ensembles and probabilistic forecasts. *Weather* **81**, 82–88 (2026).
24. Palmer, T. The ECMWF ensemble prediction system: looking back (more than) 25 years and projecting forward 25 years. *Q. J. R. Meteorol. Soc.* **145**, 12–24 (2019).
25. Price, I. et al. Probabilistic weather forecasting with machine learning. *Nature* **637**, 84–90 https://doi.org/10.1038/s41586-024-08252-9 (2025).
26. Stephens, G. L. et al. Dreary state of precipitation in global models. *J. Geophys. Res. Atmos.* **115**, D24211 (2010).
27. Mouatadid, S. et al. Adaptive bias correction for improved subseasonal forecasting. *Nat. Commun.* https://doi.org/10.1038/s41467-023-38874-y (2023).
28. Weigel, A. P., Liniger, M. A. & Appenzeller, C. The discrete brier and ranked probability skill scores. *Mon. Weather Rev.* **135**, 118–124 https://doi.org/10.1175/MWR3280.1 (2007).
29. Bouallègue, Z. B. et al. Improving medium-range ensemble weather forecasts with hierarchical ensemble transformers. *Artif. Intell. Earth Syst.* **3**, 230027 https://doi.org/10.1175/AIES-D-23-0027.1 (2024).
30. Hersbach, H. et al. The ERA5 global reanalysis. *Q. J. R. Meteorol. Soc.* **146**, 1999–2049 https://doi.org/10.1002/qj.3803 (2020).
31. Andrade, F. M., Coelho, C. A. S. & Cavalcanti, I. F. A. Global precipitation hindcast quality assessment of the subseasonal to seasonal (S2S) prediction project models. *Clim. Dyn.* **52**, 5451–5475 (2019).
32. Domeisen, D. I. V. et al. Advances in the subseasonal prediction of extreme events: relevant case studies across the globe. *Bull. Am. Meteorol. Soc.* **103**, 1473–1501 https://doi.org/10.1175/BAMS-D-20-0221.1 (2022).
33. *Section 5.3.2 SUBS-M-climate, the Sub-aseasonal Model Climate* https://confluence.ecmwf.int/spaces/FUG/pages/673550755/Section+5.3.2+SUBS-M-climate+the+sub-seasonal+model+climate (ECMWF, accessed 2022).
34. Van Schaeybroeck, B. & Vannitsem, S. Ensemble post-processing using member-by-member approaches: theoretical aspects. *Q. J. R. Meteorol. Soc.* **141**, 807–818 (2015).
35. Vannitsem, S. & Demaeyer, J. Statistical post-processing of ensemble forecasts at the Belgian Met service. *ECMWF Newsl.* **164**, 28–33 https://doi.org/10.21957/vb78ui92dj (2020).
36. Gneiting, T., Raftery, A. E., Westveld, A. H. & Goldman, T. Calibrated probabilistic forecasting using ensemble model output statistics and minimum CRPS estimation. *Mon. Weather Rev.* **133**, 1098–1118 https://doi.org/10.1175/MWR2904.1 (2005).
37. Li, W., Pan, B., Xia, J. & Duan, Q. Convolutional neural network-based statistical post-processing of ensemble precipitation forecasts. *J. Hydrol.* **605**, 127301 https://doi.org/10.1016/j.jhydrol.2021.127301 (2022).
38. Finn, T.S. Self-attentive ensemble transformer: representing ensemble interactions in neural networks for Earth System Models. Preprint at https://doi.org/10.48550/arXiv.2106.13924 (2021).
39. Scheuerer, M., Switanek, M. B., Worsnop, R. P. & Hamill, T. M. Using artificial neural networks for generating probabilistic subseasonal precipitation forecasts over California. *Mon. Weather Rev.* **148**, 3489–3506 https://doi.org/10.1175/MWR-D-20-0096.1 (2020).
40. Vitart, F. et al. The WWRP/WCRP S2S project and its achievements. *Bull. Am. Meteorol. Soc.* **106**, 791–808 https://doi.org/10.1175/BAMS-D-24-0047.1 (2025).
41. *AI Weather Quest Leaderboards* https://aiweatherquest.ecmwf.int/leaderboards/ (ECMWF, accessed 15 April 2026).
42. Talib, J., Loegel, O., Vitart, F., Hoffmann, J. & Schloer, J. The AI Weather Quest: spotlighting the best of machine learning sub-seasonal prediction. *ECMWF Sci. Blog* https://www.ecmwf.int/en/about/media-centre/science-blog/2026/ai-weather-quest-2026 (2026).
43. Oliveira, A. Northeast set to be blasted by 'most extreme cold on Earth' before Christmas. *New York Post* https://nypost.com/2025/12/01/us-news/northeast-set-to-be-blasted-by-most-extreme-cold-on-earth-before-christmas/ (2025).
44. Atienza, J. Polar vortex blankets eastern US with dangerously cold conditions, affecting over 70M Americans. Originally published by FOX Weather. *New York Post* https://nypost.com/2025/12/15/us-news/polar-vortex-blankets-eastern-us-with-dangerous-temperatures-affecting-over-70m-americans/ (2026).
45. Diffenbaugh, N. S., Pal, J. S., Trapp, R. J. & Giorgi, F. Fine-scale processes regulate the response of extreme events to global climate change. *Proc. Natl Acad. Sci. USA* **102**, 15774–15778 https://doi.org/10.1073/pnas.0506042102 (2005).
46. Masante, D. et al. *Multi-Hazard Early Warning System Global Disaster Alert and Coordination System (GDACS)—Methods for Anticipation and Rapid Impact Estimation User Manual (version 2)* https://doi.org/10.2760/1461943 (Publications Office of the European Union, 2025).
47. Kilavi, M. et al. Extreme rainfall and flooding over Central Kenya including Nairobi City during the long-rains season 2018: causes, predictability, and potential for early warning and actions. *Atmosphere* **9**, 472 https://doi.org/10.3390/atmos9120472 (2018).
48. Kolstad, E. W., Barimalala, R. & Parker, D. J. Skilful subseasonal forecasts of dry spells: a case study for Malawi. *Clim. Dyn.* **63**, 425 (2025).
49. Mwangi, E., MacLeod, D., Kniveton, D. & Todd, M. C. Variability of rainy season onsets over East Africa. *Int. J. Climatol.* **44**, 3357–3379 https://doi.org/10.1002/joc.8528 (2024).
50. Turner, W. et al. Investigating the performance of high-resolution subseasonal precipitation forecasts in support of food insecurity early warning. *Environ. Res. Clim.* **4**, 025002 https://doi.org/10.1088/2752-5295/adcadf (2025).

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Methods

Probabilistic subseasonal forecasting

We begin with a formal description of the probabilistic subseasonal forecasting problem, following the conventions of the ECMWF 2025 AI Weather Quest competition[18]. For each week-long period beginning on a date t, let $\mathbf{y}_t \in \mathbb{R}^G$ represent a vector of ground-truth weather measurements $y_{t,g}$ for each grid cell g in a global latitude–longitude grid. We used 1979–2026 ERA5 global reanalysis[30] regridded to 1.5° × 1.5° latitude–longitude resolution as our measure of ground truth and focus on predicting average 2-m temperature, total precipitation and average mean sea-level pressure across each week-long period. Both temperature and precipitation are evaluated only over land-dominated regions (that is, grid cells with at least 50% land coverage).

For each grid cell g, our aim is to predict a distribution over weather at time t, expressed as a CDF $F_{t,g}$ over $K = 5$ historical climatological quantile bins. To this end, we set $q_{t,g}(K) = \infty$ and, for each $k < K$, define the kth climatological quintile $q_{t,g}(k)$ for (t,g) as the $\frac{k}{K}$th quantile of the observations $y_{s,g}$ falling 0 days, 2 days or 4 days away from the target month–day combination in each of the previous 20 years. For each k, the probabilistic forecast $F_{t,g}(k)$ represents the predicted probability that the measurement $y_{t,g}$ will fall below the kth quintile threshold $q_{t,g}(k)$.

Given a test set $\mathcal{T}^\star$ of target dates, we measure the global quality of these probabilistic forecasts by means of the RPSS[28],

$$\mathrm{RPSS} = 1 - \frac{\sum_{t\in\mathcal{T}^\star}\sum_{g=1}^{G} w_g \,\mathrm{RPS}_{t,g}}{\sum_{t\in\mathcal{T}^\star}\sum_{g=1}^{G} w_g \,\overline{\mathrm{RPS}}_{t,g}} \quad \text{for}$$

$$\mathrm{RPS}_{t,g} = \sum_{k=1}^{K} \left(F_{t,g}(k) - O_{t,g}(k)\right)^2 \text{ and}$$

$$\overline{\mathrm{RPS}}_{t,g} = \sum_{k=1}^{K} \left(\frac{k}{K} - O_{t,g}(k)\right)^2,$$

where $O_{t,g}(k) = \mathbb{I}[y_{t,g} < q_{t,g}(k)]$ is the observed indicator of whether $y_{t,g}$ falls below the kth quintile threshold, $w_g = \cos(\text{latitude}(g))$ adjusts for the relative size of each grid cell, $\mathrm{RPS}_{t,g}$ is termed the ranked probability score (RPS) of the forecast and $\overline{\mathrm{RPS}}_{t,g}$ is the RPS of a baseline climatological forecast that assigns equal probability to each quintile bin. For precipitation, arid grid cells g with no climatological precipitation (that is, with $q_{t,g}(K-1) = 0$) are excluded from this evaluation.

Similarly, for a given grid cell g, we measure forecast quality over a test set $\mathcal{T}^\star$ by means of the spatial RPSS

$$\text{Spatial RPSS}_g = 1 - \frac{\sum_{t\in\mathcal{T}^\star} \mathrm{RPS}_{t,g}}{\sum_{t\in\mathcal{T}^\star} \overline{\mathrm{RPS}}_{t,g}}.$$

Throughout, we evaluate RPSS at two subseasonal forecasting horizons, days 19–25 (week 3) and days 26–32 (week 4). Unless otherwise noted, we use all non-leap-day Monday and Friday target dates in the years 2016–2024 as our default test set.

Baselines

Extended Data Table 1 provides a summary of the dynamical, AI, hybrid (dynamical + AI) and post-processing baselines compared with PBC in this work. Here we detail the processing steps taken to evaluate each baseline.

Raw ECMWF. At each subseasonal lead time ℓ, the operational dynamical model of ECMWF[17] outputs an ensemble of deterministic forecasts, $(\hat{y}_{m,\ell,t,g})_{m=1}^{M}$, each representing a possible future weather state. For each target weather variable and subseasonal horizon, we downloaded all weekly-aggregated ECMWF ensemble forecasts, spanning the years 2015–2026, from the subseasonal to seasonal (S2S) prediction project database[54] through the International Research Institute for Climate and Society Data Library (IRIDL)[51]. Forecasts with $M = 51$ members (1 control and 50 perturbed) were issued every Monday and Thursday from 14 May 2015 to 26 June 2023 and forecasts with $M = 101$ members were issued daily starting on 28 June 2023. We converted each ensemble into a CDF forecast $F_{\ell,t,g}(k) = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}[\hat{y}_{m,\ell,t,g} < q_{t,g}(k)]$ using the climatological quintile thresholds $q_{t,g}$.

Debiased ECMWF. To form the debiased ECMWF baseline, we followed the operational post-processing protocol of ECMWF[33] for each weather variable and subseasonal horizon. First, we downloaded all weekly-aggregated ECMWF ensemble hindcasts from the S2S database[52] through the IRIDL[51]. Each ensemble hindcast consists of $M = 11$ members, $(\hat{y}_{m,\ell,\delta,t,g})_{m=1}^{M}$, indexed by a forecast target date t and the number of years $\delta \in \{1, \ldots, 20\}$ between the forecast and corresponding hindcast target dates. Next, for each forecast target date t, we identified the set $\mathcal{S}_t$ of model climatology dates: the closest date no later than t with available hindcasts along with the next four and previous four available dates. We then formed model-based quintile thresholds for each forecast target date t, lead time ℓ and grid cell g by setting $q^{\mathrm{ECMWF}}_{\ell,t,g}(K) = \infty$ and, for each $k < K$, defining the kth model quintile $q^{\mathrm{ecmwf}}_{\ell,t,g}(k)$ as the $\frac{k}{K}$th quantile of all hindcasts $\hat{y}_{m,\ell,\delta,s,g}$ with $s \in \mathcal{S}_t$. Finally, we constructed the debiased ECMWF CDF forecast $F_{\ell,t,g}(k) = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}[\hat{y}_{m,\ell,t,g} < q^{\mathrm{ECMWF}}_{\ell,t,g}(k)]$.

FuXi-S2S. For each target weather variable and subseasonal horizon, we downloaded 51-member ensemble forecasts, spanning the test years 2017–2021, and 51-member ensemble hindcasts, spanning the training years 2002–2016, from the official FuXi-S2S forecast dataset[53]. Forecasts were subsequently aggregated to a weekly resolution, with precipitation accumulated and temperature and mean sea-level pressure averaged. Following ref. 13, we form model-based quintile thresholds for each forecast target date t, lead time ℓ and grid cell g by setting $q^{\mathrm{FuXi}}_{\ell,t,g}(K) = \infty$ and, for each $k < K$, defining the kth model quintile $q^{\mathrm{FuXi}}_{\ell,t,g}(k)$ as the $\frac{k}{K}$th quantile of all hindcasts matching the target month–day combination. Finally, we constructed the CDF forecast $F_{\ell,t,g}(k) = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}[\hat{y}_{m,\ell,t,g} < q^{\mathrm{FuXi}}_{\ell,t,g}(k)]$ from each FuXi-S2S ensemble forecast $(\hat{y}_{m,\ell,t,g})_{m=1}^{M}$ with $M = 51$.

Because FuXi-S2S and ECMWF historically issued their forecasts on different schedules, we evaluated all available forecasts from FuXi-S2S and PBC-ECMWF with target dates in the period 2017–2021. This corresponds to all Monday and Friday target dates for PBC-ECMWF and, for FuXi-S2S, all Tuesday and Friday target dates in 2017, all Wednesday and Saturday target dates in 2018, all Thursday and Sunday target dates in 2019, all Monday and Friday target dates from 3 January 2020 to 24 February 2020, all Tuesday and Saturday target dates from 29 February 2020 to 29 December 2020 and all Wednesday and Sunday target dates from 3 January 2021 to 26 December 2021.

AIFS-SUBS. For each target weather variable, subseasonal horizon and odd issuance date in 2025 (except for the 31st of any month), we generated weekly-aggregated 100-member ensemble forecasts using the AIFS-SUBS model[16]. AIFS-SUBS follows the architecture introduced in ref. 54, combining a transformer-based graph neural network encoder and decoder with a sliding-window transformer processor in which attention is computed along spiral longitudinal bands. Ensemble members are generated by conditioning the processor on independently sampled noise, injected by means of conditional layer normalization in each processor layer and the model is trained with the fair continuous RPS loss computed over four ensemble members against ERA5 targets[55]. AIFS-SUBS takes two consecutive states (t − 24 h, t) as input and predicts the atmospheric state at t + 24 h. Inputs and outputs are represented on the reduced Gaussian o96 grid (approximately 1° resolution). The variant of AIFS-SUBS used in this work includes a representation of the stratosphere and was trained exclusively on ERA5 over the periods 1979–2006 and 2012–2024, leaving a 5-year validation period.

Following ref. 16, for each forecast target date t, lead time ℓ and grid cell g, we generated 10-member ensemble hindcasts for the corresponding month–day combination in each of the previous 20 years, set $q^{\mathrm{AIFS}}_{\ell,t,g}(K) = \infty$ and, for each $k < K$, defined the kth model quintile $q^{\mathrm{AIFS}}_{\ell,t,g}(k)$ as the $\frac{k}{K}$th quantile of the aforementioned hindcasts. Finally, we constructed the CDF forecast $F_{\ell,t,g}(k) = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}[\hat{y}_{m,\ell,t,g} < q^{\mathrm{AIFS}}_{\ell,t,g}(k)]$ from each AIFS ensemble forecast $(\hat{y}_{m,\ell,t,g})_{m=1}^{M}$ with $M = 100$.

POET. The POET method[29] is a member-by-member post-processing technique that has been shown to significantly improve the medium-range weather forecasting skill of ECMWF dynamical forecasts and to outperform other statistical and AI-based post-processing techniques, including the member-by-member model output statistics post-processing technique[34] used operationally at the Royal Meteorological Institute of Belgium[35], ensemble model output statistics[36], LeNet neural network post-processing[37] and pure transformer-based post-processing[38]. POET includes a self-attention mechanism across the ensemble member dimension in its encoder-decoder convolutional U-Net architecture, allowing the model to use context from the full input forecast distribution to minimize probabilistic forecasting error. We extended POET to subseasonal horizons by making a few minor modifications to its data-processing pipeline and architecture.

First, we applied POET to weekly aggregates of weather variables rather than the instantaneous values appropriate for medium-range forecasting. As input features, we used all available variables from the S2S database[52] and a collection of 36 constant fields from ERA5[30], including orography, a land–sea mask and soil and vegetation types. Following common practice for feature embeddings, metadata, including the constant fields, lead time and weekly-averaged top-of-atmosphere incoming solar radiation, were embedded with a learned convolutional layer and used to update the highest-level features additively. To accommodate the larger set of input and output features, the model size was increased to approximately 27 million trainable parameters by increasing the number of latent features and the number of ensemble transformer layers.

POET is configured by default to predict several targets, including geopotential heights, wind and humidity on a few pressure levels and surface winds and humidity, in addition to the target variables temperature, mean sea-level pressure and precipitation. Although predicting several targets improves consistency between variables, we found that the single-target model produced higher-quality outputs for precipitation on our validation set. Hence, the final results reported for POET are from the multi-target model for temperature and pressure and from the single-target model for precipitation. The POET models were developed using the 20 years of 10-member ECMWF hindcasts issued in 2024, with the hindcasts for the years 2004–2022 used for training and the 2023 hindcasts used for validation.

For each target weather variable, subseasonal horizon and ECMWF forecast issuance date in 2024 (every Monday and Thursday to 21 November and every odd date thereafter), we generated weekly-aggregated ensemble forecasts, $(\hat{y}_{m,\ell,t,g})_{m=1}^{M}$, by applying POET to the $M = 100$ perturbed ECMWF ensemble members. We then converted each ensemble into a CDF forecast $F_{\ell,t,g}(k) = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}[\hat{y}_{m,\ell,t,g} < q_{t,g}(k)]$ using the climatological quintile thresholds $q_{t,g}$.

PBC

PBC (Fig. 1) corrects the systematic errors in probabilistic forecasts by training and applying two complementary machine learning corrections (Debias++ and Persistence++), projecting the output of each onto the space of valid distributions and finally averaging the resulting distributions. For a given weather variable, target date $t^\star$ and lead time $\ell^\star$, each machine learning algorithm is given access to a model's uncorrected CDF forecasts $\mathbf{F}_{\ell^\star,t^\star} = (F_{\ell^\star,t^\star,g})_{g=1}^{G}$ for the target, a dataset of historical CDF forecasts $\mathbf{F}_{\ell,t} = (F_{\ell,t,g})_{g=1}^{G}$ from the model with training set target dates $t \in \mathcal{T}$ and lead times $\ell \in \mathcal{L}$ and observed cumulative quantile bin indicators $\mathbf{O}_t = (O_{t,g})_{g=1}^{G}$ for each $t \in \mathcal{T}$. To mimic operational deployment, each algorithm is only given access to training observations that would have been fully observable $\ell^\star - 1$ days previous to the target date $t^\star$.

Debias++. For each quantile bin k, Debias++ (Algorithm 1) performs probabilistic forecast correction in four steps. First, to account for heterogeneity in forecast errors over time, Debias++ adaptively selects a span s of observations around the target day of year for probabilistic debiasing and a range $d^\star$ of forecast issuance dates for ensembling. Second, Debias++ forms ensemble mean probabilistic forecasts $\bar{\mathbf{F}}_t(k)$ by averaging its input CDF forecasts over the selected range of issuance dates. Third, to directly counteract systematic errors, Debias++ corrects the target CDF forecast $\bar{\mathbf{F}}_{t^\star}(k)$ by subtracting the probabilistic prediction bias—the average forecast value $\bar{\mathbf{F}}_t(k)$ minus the average observed value $\mathbf{O}_t(k)$—over the selected debiasing window across the previous Y years. Finally, the results are projected into the valid probability range $[0, 1]$.

In every application of Debias++, we set $Y = 20$, selected the configuration $(s, d^\star) \in \{(14, 1), (28, 1), (35, 1)\}$ that minimized mean RPS over the 3 years preceding the target date $t^\star$ and converted each available ensemble forecast $(\hat{y}_{m,\ell,t,g})_{m=1}^{M}$ and hindcast $(\hat{y}_{m,\ell,\delta,t,g})_{m=1}^{M}$ into an input CDF forecast using the observed climatological quintile thresholds $q_{t,g}$:

$$F_{\ell,t,g}(k) = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}[\hat{y}_{m,\ell,t,g} < q_{t,g}(k)] \text{ and}$$

$$F_{\ell,t-\delta\,\mathrm{years},g}(k) = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}[\hat{y}_{m,\ell,\delta,t,g} < q_{t-\delta\,\mathrm{years},g}(k)].$$

Algorithm 1. Debias++

input target date $t^\star$; quantile bin k; lead time $\ell^\star$; span s; # issuance dates $d^\star$; # training years Y; training set of observed cumulative quantile bin indicators $(\mathbf{O}_t)_{t\in\mathcal{T}}$ and cumulative probability forecasts $(\mathbf{F}_{\ell,t})_{\ell\in\mathcal{L},t\in\mathcal{T}\cup\{t^\star\}}$

initialize days per year $D = 365.242199$

$$\mathcal{S} = \left\{t \in \mathcal{T}\colon \mathrm{year_diff} := \left\lfloor \frac{t^\star - t}{D} \right\rfloor \le Y \text{ and } \mathrm{day_diff} := \frac{365}{2} - \left| \lfloor (t^\star - t) \bmod D \rfloor - \frac{365}{2} \right| \le s\right\}$$

// Form ensemble mean probabilistic forecasts across issuance dates

for target and training dates $t \in \mathcal{S} \cup \{t^\star\}$ **do**

$\bar{\mathbf{F}}_t(k) \leftarrow \mathrm{mean}((\mathbf{F}_{\ell^\star-d+1,t}(k))_{1\le d\le d^\star})$

output $\mathrm{clip}\,(\bar{\mathbf{F}}_{t^\star}(k) + \mathrm{mean}((\mathbf{O}_t(k) - \bar{\mathbf{F}}_t(k))_{t\in\mathcal{S}}), [0,1])$

Persistence++. For each quantile bin k and grid cell g, Persistence++ (Algorithm 2) accounts for both recent weather trends and climatological shifts by regressing the observed indicator $O_{t,g}(k)$ onto the historical model forecast $F_{\ell^\star,t,g}(k)$, two recent and fully observable lagged indicators $O_{t-\ell^\star+6,g}(k)$ and $O_{t-2\ell^\star+5,g}(k)$ and a probabilistic climatology formed from the preceding 20 years of observed indicators with matching month and day. In the climatology computation, any missing indicators are imputed with the prior value $\frac{k}{K}$. The learned regression weights are applied to the target forecast $F_{\ell^\star,t^\star,g}(k)$, lagged observations and climatology to produce a corrected output that is then projected into the valid probability range $[0, 1]$.

When applying Persistence++ to POET forecasts or ECMWF precipitation forecasts, we again converted each available ensemble forecast and hindcast into an input CDF forecast using the observed climatological quintile thresholds. In all other applications of Persistence++, we used model-based quintile thresholds $q^{\mathrm{model}}_{\ell,t,g}$ for a given forecast target date t to convert both the ensemble forecast $(\hat{y}_{m,\ell,t,g})_{m=1}^{M}$ and the associated ensemble hindcasts $(\hat{y}_{m,\ell,\delta,t,g})_{m=1}^{M}$ into input CDF forecasts:

$$F_{\ell,t,g}(k) = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}\,[\hat{y}_{m,\ell,t,g} < q^{\mathrm{model}}_{\ell,t,g}(k)] \text{ and}$$

$$F_{\ell,t-\delta\,\mathrm{years},g}(k) = \frac{1}{M}\sum_{m=1}^{M} \mathbb{I}\,[\hat{y}_{m,\ell,\delta,t,g} < q^{\mathrm{model}}_{\ell,t,g}(k)].$$

Algorithm 2. Persistence++

input target date $t^\star$; quantile bin k; lead time $\ell^\star$; training set of observed cumulative quantile bin indicators $(\mathbf{O}_t)_{t\in\mathcal{T}}$ and cumulative probability forecasts $(\mathbf{F}_{\ell^\star,t})_{t\in\mathcal{T}\cup\{t^\star\}}$

initialize forecast period length $L=7$, # climatology years $Y=20$

// Form rolling probabilistic climatology from observed indicators

for target and training dates $t\in\mathcal{T}\cup\{t^\star\}$ **do**

$\mathbf{C}_t(k)=\text{mean}(\{\mathbf{O}_s(k):\text{monthday}(s)=\text{monthday}(t),\ 1\le\text{year}(t)-\text{year}(s)\le Y\})$

// Combine probabilistic forecast, climatology and lagged observations

for grid points $g=1$ to G **do**

$$\widehat{\boldsymbol{\beta}}_g\in\operatorname{argmin}_{\boldsymbol{\beta}}\sum_{t\in\mathcal{T}}\left(O_{t,g}(k)-\boldsymbol{\beta}^\top[1,C_{t,g}(k),O_{t-\ell^\star-L+1,g}(k),O_{t-2\ell^\star-L+2,g}(k),F_{\ell^\star,t,g}(k)]\right)^2$$

$$\textbf{output}\ \text{clip}\left((\widehat{\boldsymbol{\beta}}_g^\top[1,C_{t^\star,g}(k),O_{t^\star-\ell^\star-L+1,g}(k),O_{t^\star-2\ell^\star-L+2,g}(k),F_{\ell^\star,t^\star,g}(k)])_{g=1}^G,[0,1]\right)$$

Projection and ensembling. It is possible for the quantile bin predictions of Debias++ and Persistence++ to violate the constraint that cumulative probabilities must be non-decreasing across quantile thresholds. In such cases, we use isotonic regression[56] to project their output forecasts onto the space of valid CDFs. This operation is guaranteed to improve forecast skill as, by minimizing squared error to the set of valid CDFs, it increases RPSS for all possible observations simultaneously. The final PBC forecast averages the two projected corrections to capture the complementary benefits of each.

MicroDuet. To form the MicroDuet model used in the AI Weather Quest, we considered three combinations of the PBC-ECMWF and PBC-POET forecasts—an equal-weighted average, full weight on PBC-ECMWF and full weight on PBC-POET—and selected the combination that maximized 2024 RPSS for each variable and lead time. The resulting selection gave equal weight to PBC-ECMWF and PBC-POET for temperature, full weight to PBC-ECMWF for precipitation and full weight to PBC-POET for mean sea-level pressure.

AI Weather Quest evaluation

For each target weather variable and model, Fig. 4 reports the AI Weather Quest leaderboard scores for the September–October–November contest season[41]. The Quest scored each model against ERA5-T ground truth[30] using the RPSS across all Monday target dates $\mathcal{T}^\star$ from 1 September 2025 to 24 November 2025 (inclusive) for week 3 and from 8 September 2025 to 1 December 2025 (inclusive) for week 4.

Forecasting extreme events

When forecasting extreme highs or extreme lows, each model is used to predict the probability of the top semidecile (measurements verifying above the 95th percentile) or the bottom semidecile (measurements verifying below the 5th percentile) respectively. Given a test set $\mathcal{T}^\star$ of target dates, we measured extreme forecast quality by means of the BSS[57],

$$\text{BSS}=1-\frac{\sum_{t\in\mathcal{T}^\star}\sum_{g=1}^G w_g\,\text{BS}_{t,g}}{\sum_{t\in\mathcal{T}^\star}\sum_{g=1}^G w_g\,\overline{\text{BS}}_{t,g}}\quad\text{for}$$
$$\text{BS}_{t,g}=(F_{t,g}(k)-O_{t,g}(k))^2\ \text{ and }\ \overline{\text{BS}}_{t,g}=\left(\frac{k}{K}-O_{t,g}(k)\right)^2, \tag{1}$$

where $K=20$, $k=1$ for extreme lows, $k=K-1$ for extreme highs, $\text{BS}_{t,g}$ is termed the Brier score (BS) of the forecast and $\overline{\text{BS}}_{t,g}$ is the BS of a baseline climatological forecast that assigns equal probability to each semidecile bin.

Flood forecasting

To assess forecast skill during observed flood events, we constructed a global catalogue of flood occurrences from the GDACS (https://www.gdacs.org). Flood events (event type FL) at all alert levels (green, orange, red) were retrieved through the GDACS REST API for the date range 1 January 2016 to 29 May 2026. Each GDACS event record provides a centroid (latitude, longitude) coordinate and an event start date; from these we derived for every event (1) a geographic bounding box of ±10° around the centroid (clamped to [−90°, 90°] in latitude) and (2) a 7-day target period beginning on the event start date. Only events with start dates matching the target date of an ECMWF dynamical forecast (that is, Monday or Friday start dates until 15 July 2023 for week 3 or until 22 July 2023 for week 4 and all start dates thereafter) were evaluated. This yielded 2,435 flood events for week 3 and 2,428 for week 4. The severe flood analysis filtered these events to floods with orange or red alert levels yielding 88 events at each lead time.

For a given test set $\mathcal{T}^\star$ of flood event target dates, we measured flood forecast quality by means of the BSS (1) for the top precipitation semidecile bin ($k=K-1$ for $K=20$) with g now enumerating the bounding box grid cells.

Statistical methods

Paired Wilcoxon signed-rank tests were used to test for significant RPSS differences across the $T=26$ weeks of the September–October–November and December–January–February AI Weather Quest seasons[42]. Because the exact issuance dates of the FuXi-S2S and ECMWF dynamical model seldom coincided in the 2017–2021 test period of Extended Data Fig. 6, we reported (two-sided) stationary block bootstrap[58] confidence intervals based on the bias-corrected and accelerated bootstrap[59] with automatic block length selection[60], 5,000 bootstrap replicates, $T=514$ target dates for FuXi-S2S and $T=522$ target dates for PBC-ECMWF.

For significance testing in Fig. 5c, we used paired one-sided bootstrap tests of BSS improvement using the bias-corrected and accelerated bootstrap with 5,000 bootstrap replicates, $T=88$ severe flood events, $T=2{,}435$ total week-3 flood events and $T=2{,}428$ total week-4 flood events. For all other significance tests, we used paired one-sided stationary block bootstrap tests of skill score improvement using the bias-corrected and accelerated bootstrap with automatic block length selection, 5,000 bootstrap replicates and T target dates for $T=939$ (Figs. 2 and 5a and Extended Data Figs. 2, 7a and 9a), $T=179$ (Fig. 3a), $T=104$ (Fig. 3b and Extended Data Fig. 7b, week 3), $T=107$ (Fig. 3b and Extended Data Fig. 7b, week 4), $T=232$ (Extended Data Fig. 1, December–January–February), $T=237$ (Extended Data Fig. 1, March–April–May), $T=236$ (Extended Data Fig. 1, June–July–August) and $T=234$ (Extended Data Fig. 1, September–October–November).

Data availability

All ERA5 data used in this study is available through the Climate Data Store of the Copernicus Climate Change Service (https://cds.climate.copernicus.eu). All dynamical data used in this study are available in the S2S prediction project database[52]. S2S is a joint initiative of the World Weather Research Programme and the World Climate Research Programme. The original S2S database is hosted at ECMWF as an extension of the TIGGE database (https://apps.ecmwf.int/datasets/data/s2s). All AI Weather Quest evaluations are available through the AI Weather Quest leaderboard (https://aiweatherquest.ecmwf.int/leaderboards/). All FuXi-S2S forecasts are available in the official FuXi-S2S forecast dataset[53]. All flood events are available through the GDACS REST API (https://www.gdacs.org). All maps were generated using the Cartopy Python package[61] with coastlines, continguous US borders and lake boundaries provided by Natural Earth (https://www.naturalearthdata.com/). Source data are provided with this paper.

Code availability

Open-source code implementing PBC and the experiments in this work is available at GitHub (https://github.com/mouatadid/pbc) and Zenodo (https://doi.org/10.5281/zenodo.22238796)[62]. Open-source code for the POET model is available at Github (https://github.com/ecmwf-lab/poet/). Code for training AIFS-SUBS is available in the anemoi open-source package (https://github.com/ecmwf/anemoi-core).

51. *ECMWF S2S data: ECMWF S2S ECMF: ECMWF Ensemble* https://iridl.ldeo.columbia.edu/SOURCES/.ECMWF/.S2S/.ECMF/ (IRI, 2026).
52. Vitart, F. et al. The subseasonal to seasonal (S2S) prediction project database. *Bull. Am. Meteorol. Soc.* **98**, 163–173 https://doi.org/10.1175/BAMS-D-16-0017.1 (2017).
53. Zhong, X. FuXi-S2S (Revision f63dc2e). *Hugging Face* https://huggingface.co/datasets/FudanFuXi/FuXi-S2S (2025).
54. Lang, S. et al. AIFS—ECMWF's data-driven forecasting system. Preprint at https://doi.org/10.48550/arXiv.2406.01465 (2024).
55. Lang, S. et al. AIFS-CRPS: ensemble forecasting using a model trained with a loss function based on the continuous ranked probability score. *NPJ Artif. Intell.* https://doi.org/10.1038/s44387-026-00073-7 (2026).
56. Best, M. J. & Chakravarti, N. Active set algorithms for isotonic regression; a unifying framework. *Math. Program.* **47**, 425–439 (1990).
57. Brier, G. W. Verification of forecasts expressed in terms of probability. *Mon. Weather Rev.* **78**, 1–3 (1950).
58. Politis, D. N. & Romano, J. P. The stationary bootstrap. *J. Am. Stat. Assoc.* **89**, 1303–1313 (1994).
59. Efron, B. & Narasimhan, B. The automatic construction of bootstrap confidence intervals. *J. Comput. Graph. Stat.* **29**, 608–619 https://doi.org/10.1080/10618600.2020.1714633 (2020).
60. Patton, A., Politis, D. N. & White, H. Correction to "Automatic block-length selection for the dependent bootstrap" by D. Politis and H. White. *Econometr. Rev.* **28**, 372–375 https://doi.org/10.1080/07474930802459016 (2009).
61. Met Office. Cartopy: A cartographic Python Library with a Matplotlib Interface https://cartopy.readthedocs.io (2010).
62. Mouatadid, S. & Mackey, L. mouatadid/pbc: Replication code for advancing subseasonal forecasting with machine learning (version v1.0.0). *Zenodo* https://doi.org/10.5281/zenodo.22238796 (2026).
63. Lu, B. et al. *CMAandFDU* (team page) https://aiweatherquest.ecmwf.int/team/cmaandfdu/ (AI Weather Quest, accessed 14 April 2026).
64. Lu, P. *LP* (team page) https://aiweatherquest.ecmwf.int/team/lp/ (AI Weather Quest, accessed 14 April 2026).

Author contributions H.G. performed her work as an intern at Microsoft Research. H.G., S.M., P.O., J.C., H.D., Z.N., G.F., A.L., J.A.W. and L.M. conceptualized the work. S.M., J.C., A.L., J.A.W. and L.M. managed the project. H.G., S.M., P.O., G.F., J.T., J.A.W. and L.M. curated data. H.G., S.M., P.O., H.D., Z.N., J.A.W. and L.M. developed models. H.G., S.M., P.O., J.B., G.F., J.T., J.S., J.A.W. and L.M. wrote code. H.G., S.M., P.O., J.B., J.S., J.A.W. and L.M. conducted experiments and evaluation. H.G., S.M., P.O., J.C. and L.M. wrote the original draft of the paper. H.D., J.B., G.F., A.L., J.S., J.T., J.A.W., J.C. and L.M. reviewed and edited the paper.

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Competing interests J.C., H.D., Z.N., J.B., A.L., J.A.W. and L.M. own Microsoft stock. H.D., Z.N., A.L., J.A.W. and L.M. are Microsoft employees. The remaining authors declare no competing interests.

Additional information
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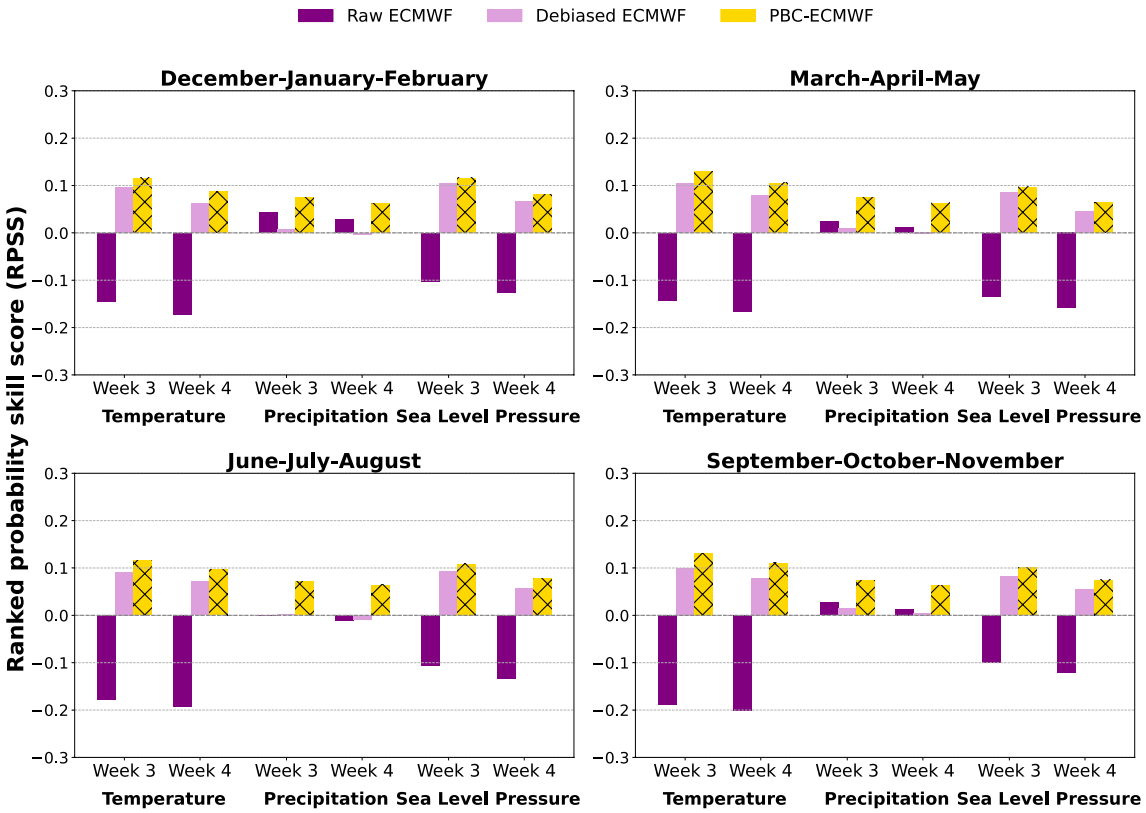
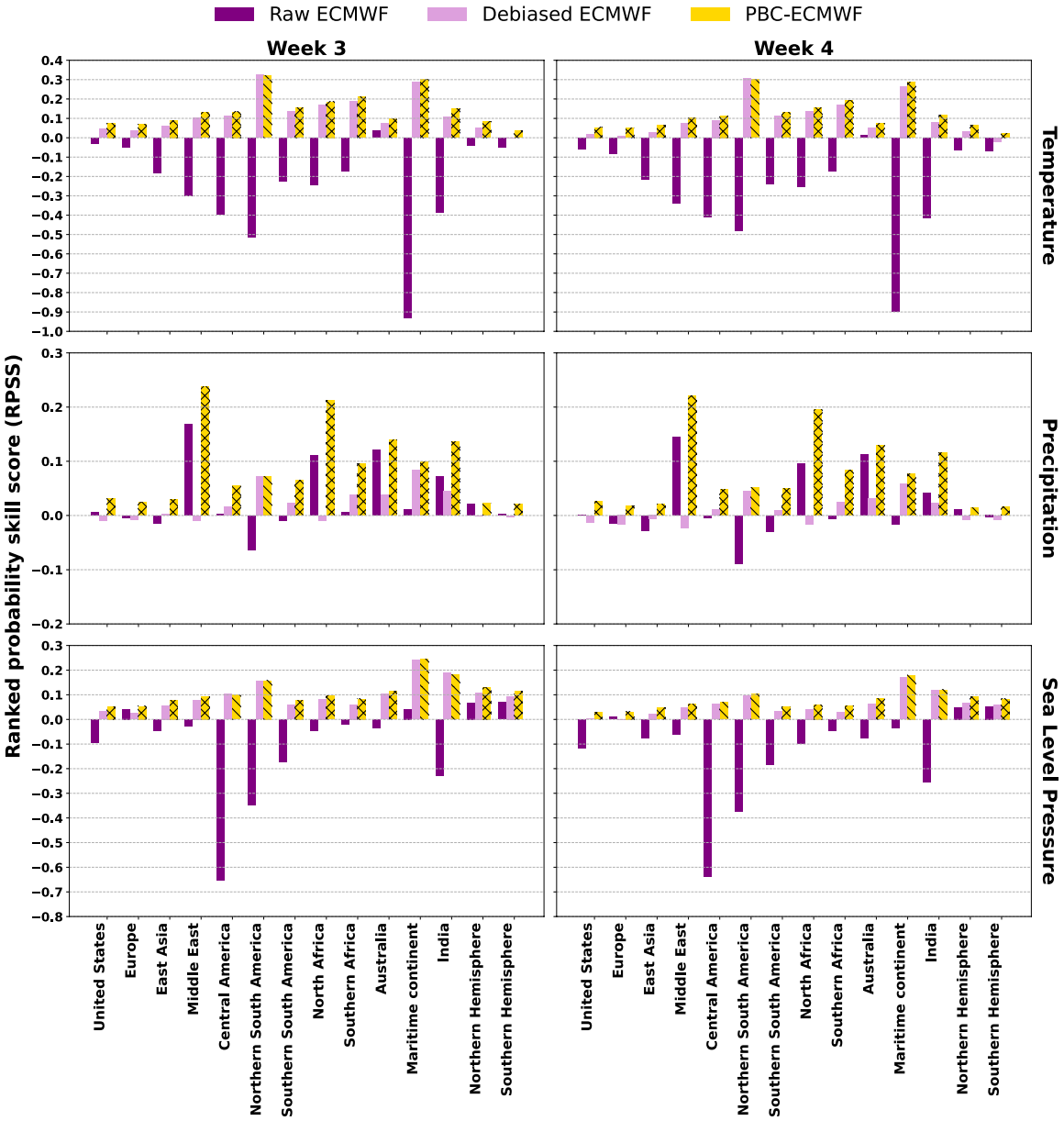
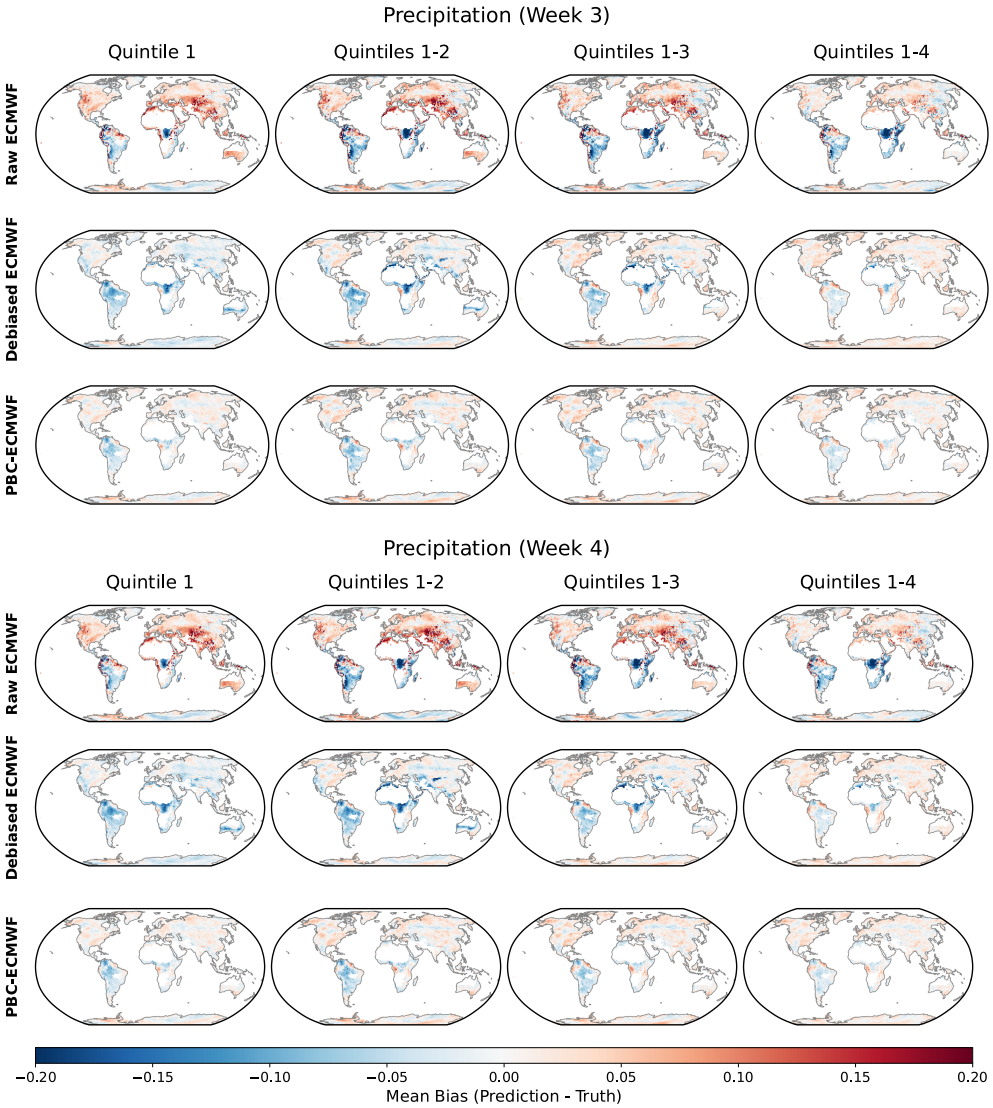


Extended Data Fig. 1 | Global forecast skill per season for the leading dynamical model (ECMWF) and its probabilistic bias correction (PBC). Across the years 2016-2024, PBC boosts the skill of ECMWF and operationally debiased ECMWF in every season. Crosshatching indicates significant improvement ($P < 0.05$) over raw and debiased ECMWF.

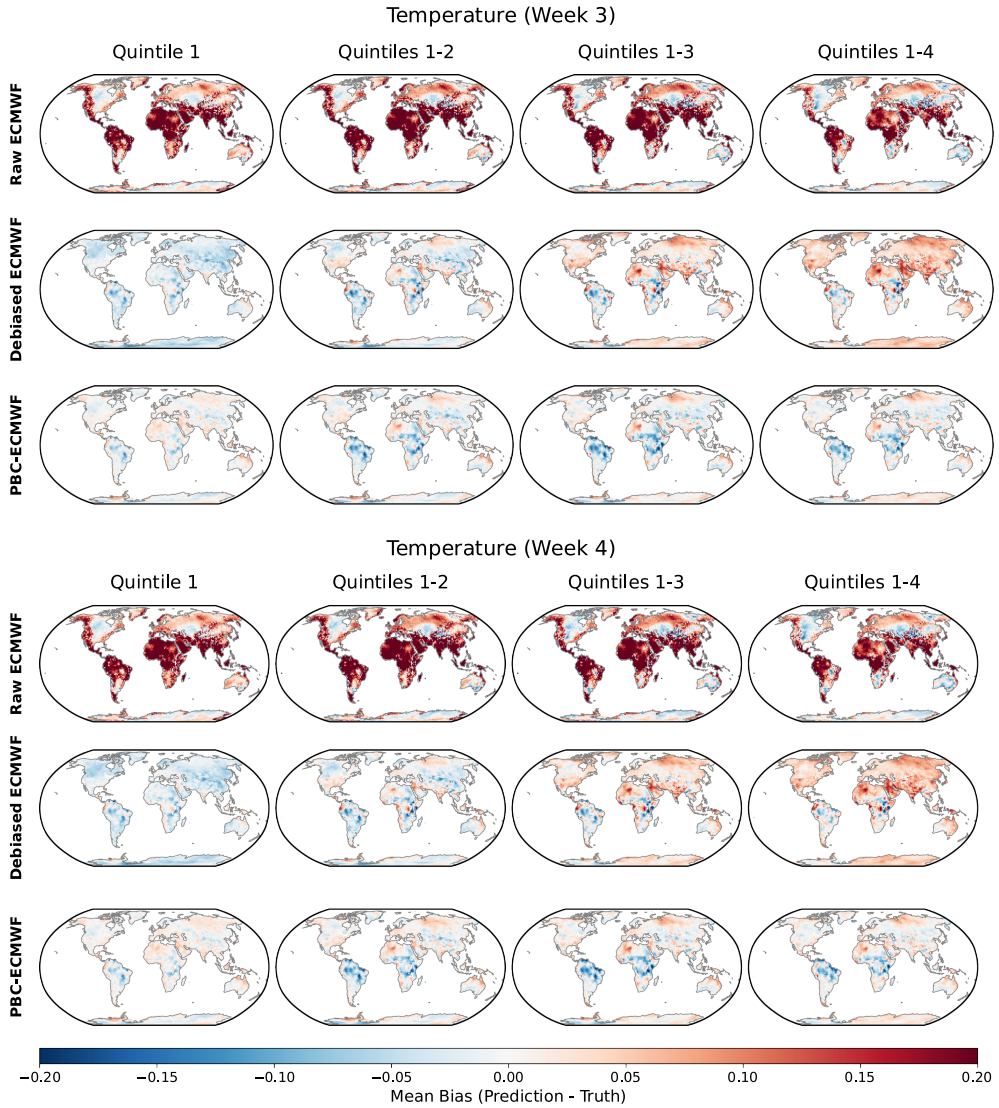


United States	26°N–50°N, 125°W–67°W	**North Africa**	0°N–40°N, 18°W–45°E
Europe	35°N–70°N, 10°W–50°E	**Southern Africa**	35°S–0°N, 10°E–40°E
East Asia	20°N–50°N, 90°E–145°E	**Australia**	45°S–10°S, 110°E–179°E
Middle East	15°N–40°N, 30°E–60°E	**Maritime continent**	10°S–20°N, 90°E–150°E
Central America	8°N–32°N, 118°W–80°W	**India**	5°N–35°N, 65°E–90°E
Northern South America	0°N–12°N, 82°W–35°W	**Northern Hemisphere**	0°N–90°N, 180°W–180°E
Southern South America	56°S–0°N, 80°W–36°W	**Southern Hemisphere**	90°S–0°N, 180°W–180°E

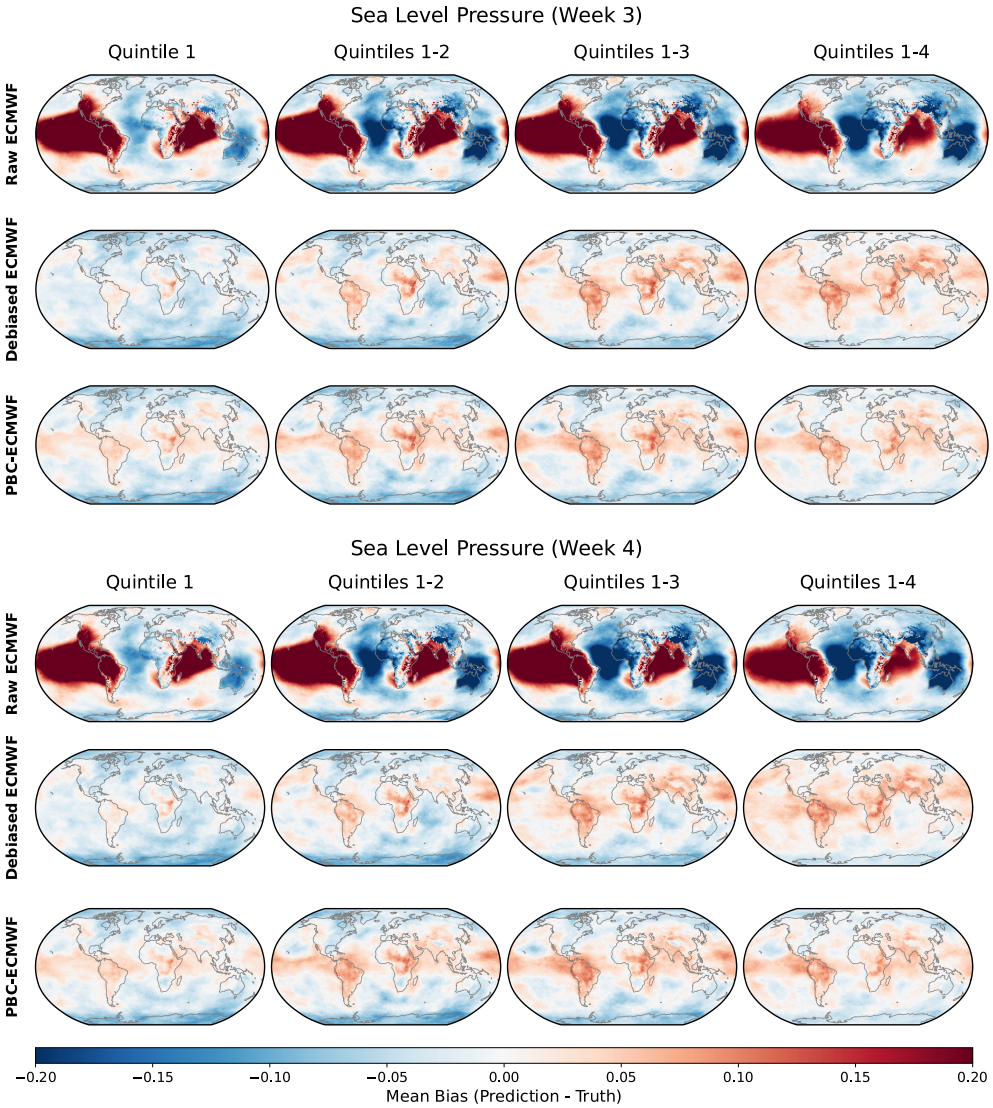
Extended Data Fig. 2 | Regional forecast skill of the leading dynamical model (ECMWF) and its probabilistic bias correction (PBC). Across the years 2016-2024 and world regions defined by the associated latitude-longitude bounding boxes, PBC boosts the skill of ECMWF and operationally debiased ECMWF. Crosshatching indicates significant improvement ($P < 0.05$) over raw and debiased ECMWF, while single hatching indicates significant improvement over one baseline.



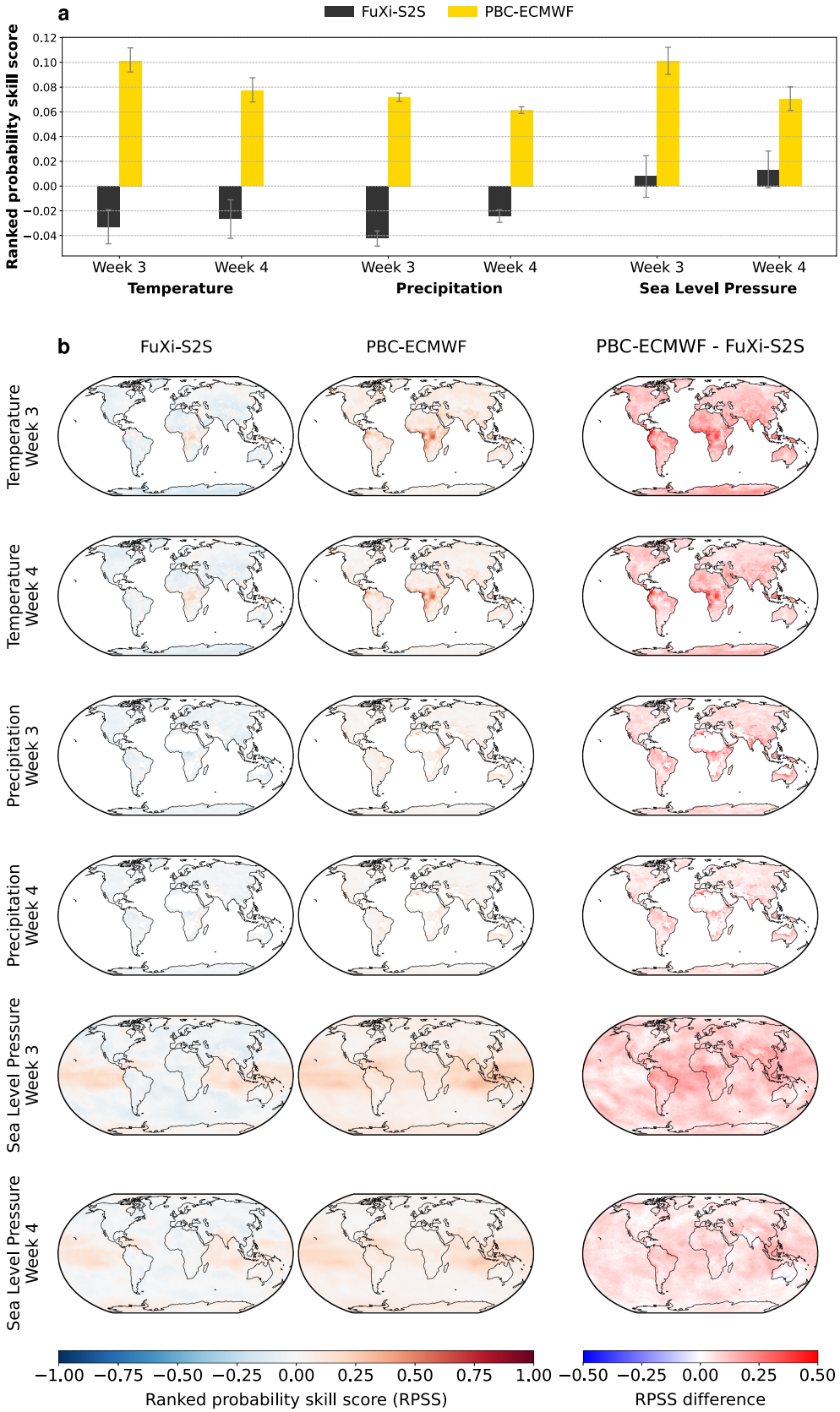
Extended Data Fig. 3 | Spatial distribution of precipitation model bias for the leading dynamical model (ECMWF) and its probabilistic bias correction (PBC). Across the globe, the years 2016–2024, and all probabilistic quintile bins, PBC reduces the systematic bias of ECMWF more effectively than operational debiasing. Coastlines from Natural Earth (https://www.naturalearthdata.com/).



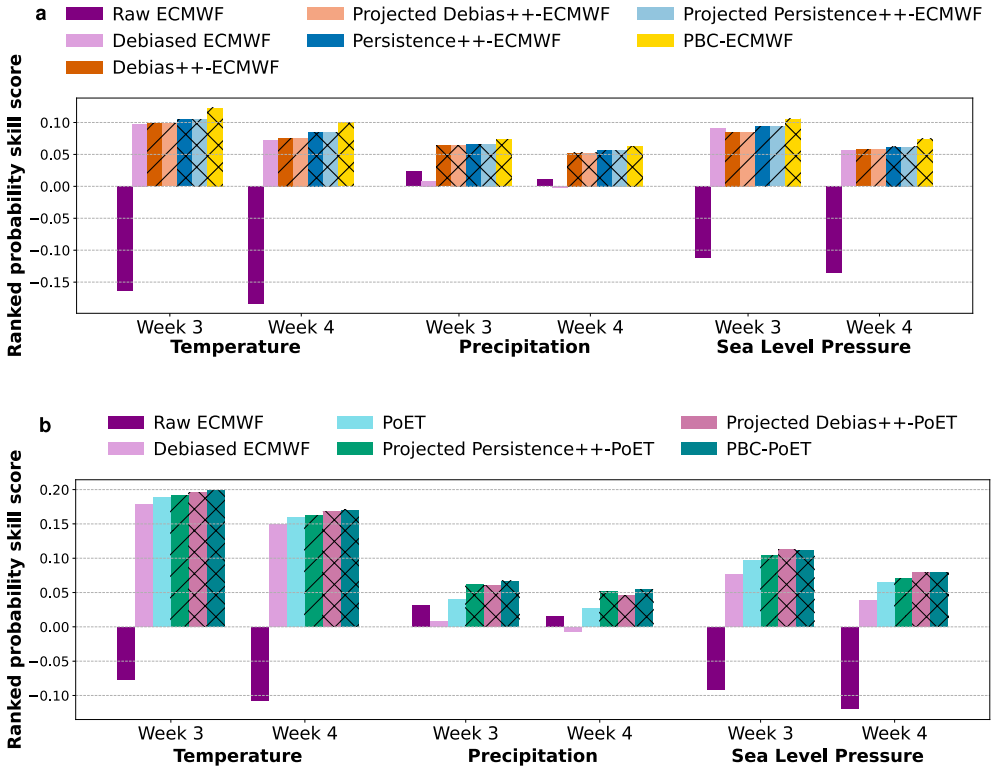
Extended Data Fig. 4 | Spatial distribution of temperature model bias for the leading dynamical model (ECMWF) and its probabilistic bias correction (PBC). Across the globe, the years 2016-2024, and all probabilistic quintile bins, PBC reduces the systematic bias of ECMWF more effectively than operational debiasing. Coastlines from Natural Earth (https://www.naturalearthdata.com/).



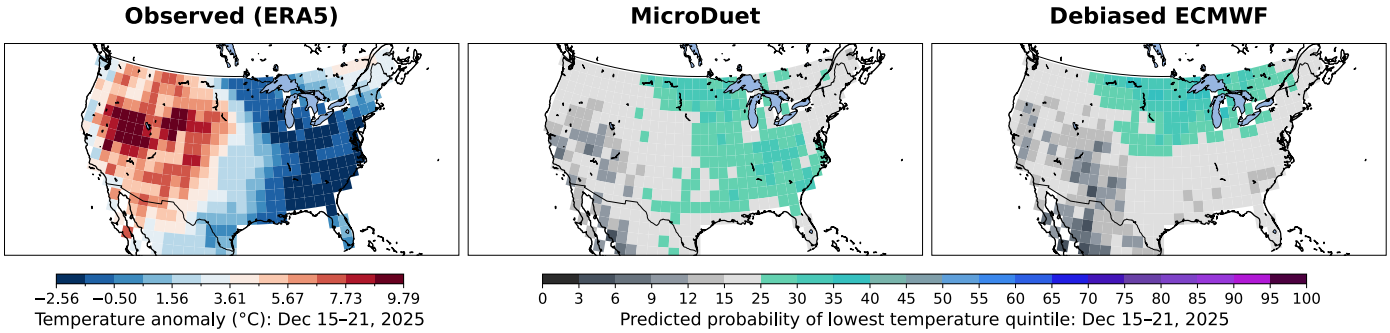
Extended Data Fig. 5 | Spatial distribution of mean sea level pressure model bias for the leading dynamical model (ECMWF) and its probabilistic bias correction (PBC). Across the globe, the years 2016–2024, and all probabilistic quintile bins, PBC substantially reduces the systematic bias of ECMWF. Coastlines from Natural Earth (https://www.naturalearthdata.com/).



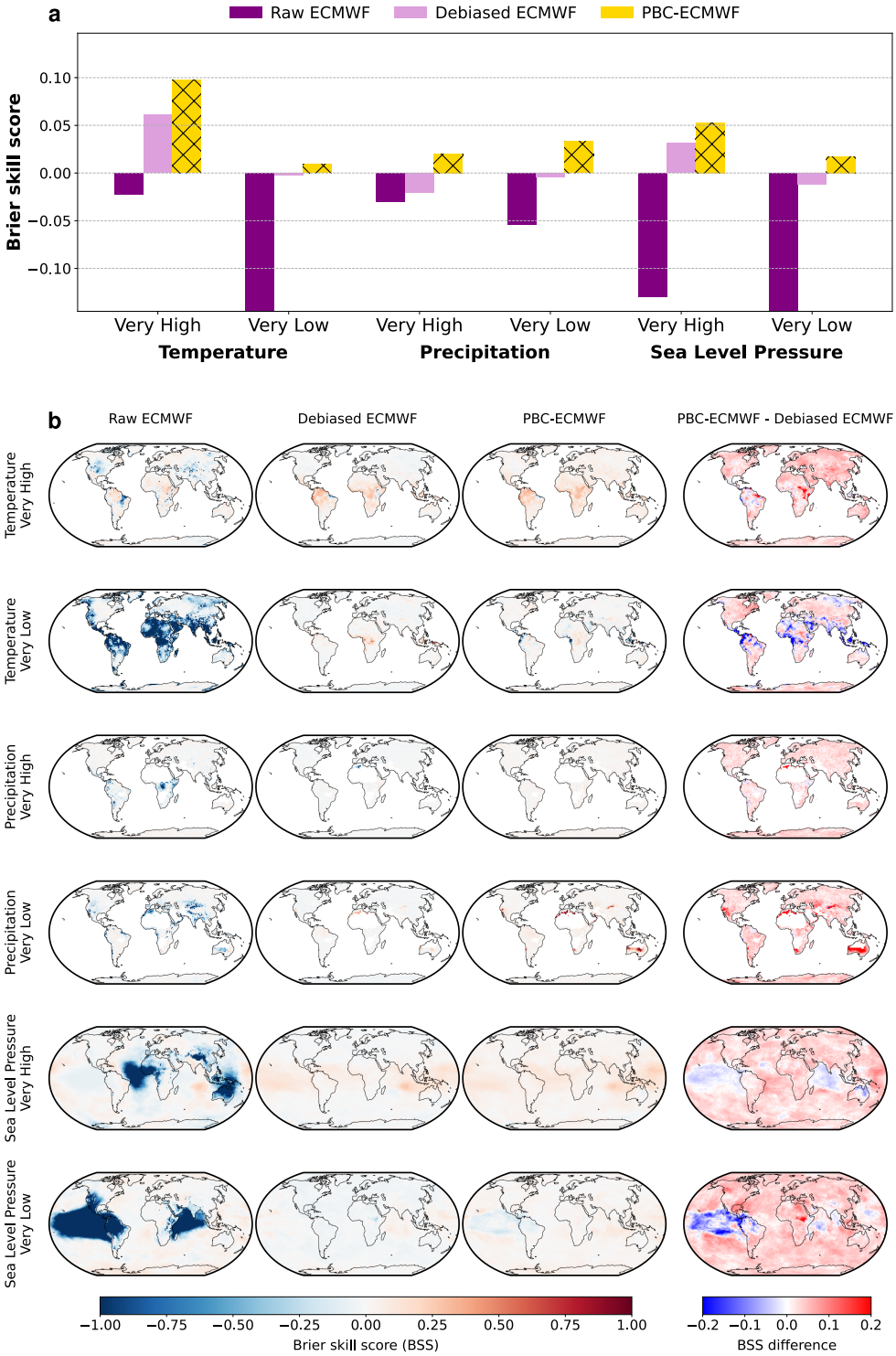
Extended Data Fig. 6 | Global forecast skill of FuXi-S2S and PBC-ECMWF. Across the years 2017–2021, PBC-ECMWF **(a)** displays a pronounced improvement in skill over the FuXi-S2S AI model for each variable and lead time with **(b)** 99–100% of grid cells showing skill gains for temperature, 97–99% for precipitation, and 96–100% for mean sea level pressure. Arid grid cells with no climatological precipitation are excluded. The error bars display 95% bootstrap confidence intervals. Coastlines in **b** from Natural Earth (https://www.naturalearthdata.com/).



Extended Data Fig. 7 | Global forecast skill of the leading dynamical model (ECMWF), the hybrid (AI + dynamical) model PoET, and the components of their probabilistic bias corrections (PBC). (**a**) Across the years 2016–2024, for every variable and lead time, Persistence++ and Debias++ individually boost the subseasonal skill of raw ECMWF forecasts and contribute to robust skill gains over operational debiasing through ensembling. (**b**) In 2024, PoET already improves the skill of its dynamical input, ECMWF. Nevertheless, PBC further enhances skill for every variable and lead time thanks to direct adaptive CDF optimization with Debias++, the integration of recent and climatological observations through Persistence++, and the robust combination of these complementary corrections via ensembling. Crosshatching indicates significant improvement ($P < 0.05$) over all baselines, while single hatching indicates significant improvement over at least one baseline.



Extended Data Fig. 8 | Forecasting the December 2025 cold air outbreak in the Eastern United States. MicroDuet's real-time week 4 temperature forecast (issued on November 20, 2025 for the AI Weather Quest competition) anticipated the mid-December cold air outbreak in the Eastern United States[43], an extreme cold event[44] missed by the leading debiased dynamical model from ECMWF. Coastlines, contiguous US borders and lakes from Natural Earth (https://www.naturalearthdata.com/).



Extended Data Fig. 9 | Forecasting extreme weather with the leading dynamical model (ECMWF) and its probabilistic bias correction (PBC) in week 4. Across the globe and the years 2016–2024, PBC boosts ECMWF extreme forecast skill more effectively than operational debiasing protocols with 96% of grid cells showing skill gains for precipitation, 90% for sea level pressure, and 85% for temperature. (**a**) Crosshatching indicates significant improvement ($P < 0.05$) over raw and debiased ECMWF. (**b**) Arid grid cells with no climatological precipitation are excluded. Coastlines from Natural Earth (https://www.naturalearthdata.com/).

Extended Data Table 1 | Baselines

Baseline	Model Type	Evaluation Periods
Raw ECMWF[17]	Dynamical	2016–2024, 2025 real-time
Debiased ECMWF[33]	Post-processing, Dynamical	2016–2024, 2025 real-time
FuXi-S2S[53]	AI	2017–2021
PoET[29]	Post-processing, Hybrid	2024
AIFS-SUBS[16]	AI	2025, 2025 real-time
FengshunHybrid[63]	Hybrid	2025 real-time
LPM[64]	Post-processing, Hybrid	2025 real-time
Debiased ECCC[52]	Post-processing, Dynamical	2025 real-time
Debiased KMA[52]	Post-processing, Dynamical	2025 real-time
Debiased JMA[52]	Post-processing, Dynamical	2025 real-time
Debiased HMCR[52]	Post-processing, Dynamical	2025 real-time
Debiased NOAA[52]	Post-processing, Dynamical	2025 real-time
Dynamical MME[52]	Post-processing, Dynamical	2025 real-time

Dynamical, AI, hybrid (dynamical + AI), and post-processing baselines compared with probabilistic bias correction in this work. References 63,64.